

Week 10 (7.24-8.2)

Module 1:

(A)Run the first HPLC to verify isoeugenol 、 vanillin production

1. Inoculation

Original strains: BW-pYB1a-*eGFP*、BW-pYB1a-*progo*、BW-pYB1a-*novIf*

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-*eGFP*、BW-pYB1a-*progo*、BW-pYB1a-*novIf*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis

BW-pYB1a-*progo* using 4-PG to catalyze the induced enzymes for isoeugenol production . BW-pYB1a-*novIf* using isoeugenol to catalyze the induced enzymes for vanillin production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀ = 6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, BW-pYB1a-*progo* resuspend the bacterial cells in pH=7.5 Tris-HCL buffer. BW-pYB1a-*novIf* resuspend the bacterial cells in pH=7.5 KPi buffer. Transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 1、3hours.

4. High-Performance Liquid Chromatography (HPLC) Detection of isoeugenol 、 vanillin

1.Preparation of the mobile phase.

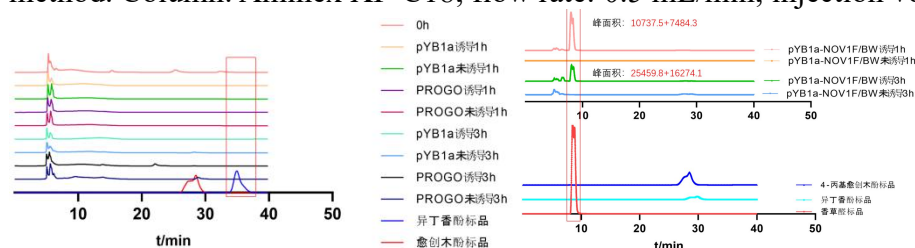
The mobile phase was prepared: methanol and 0.1% glacial acetic acid were also prepared.

2.Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

3.HPLC detection.

The samples were detected by HPLC using the methanol and 0.1% glacial acetic acid method. Column: Aminex XP-C18; flow rate: 0.5 mL/min; injection volume: 10 μ L.



(B)Run the second HPLC to verify isoeugenol 、 vanillin production

1. Inoculation

Original strains: BW-pYB1a-*eGFP*、BW-pYB1a-*progo*、BW-pYB1a-*progo-novIf*
Cultivation in a shaker at 37°C for 12 hours

2. Induction

Induction of BW-pYB1a-*eGFP*、BW-pYB1a-*progo*、BW-pYB1a-*progo-novIf*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis

BW-pYB1a-*progo* using 4-PG to catalyze the induced enzymes for isoeugenol production. BW-pYB1a-*progo-novIf* using 4-PG to catalyze the induced enzymes for vanillin production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀ = 6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, BW-pYB1a-*progo* resuspend the bacterial cells in pH=7.5 KPi buffer. BW-pYB1a-*progo-novIf* resuspend the bacterial cells in pH=7.5 KPi buffer. Transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 1、3 hours.

4.High-Performance Liquid Chromatography (HPLC) Detection of isoeugenol 、vanillin

1.Preparation of the mobile phase.

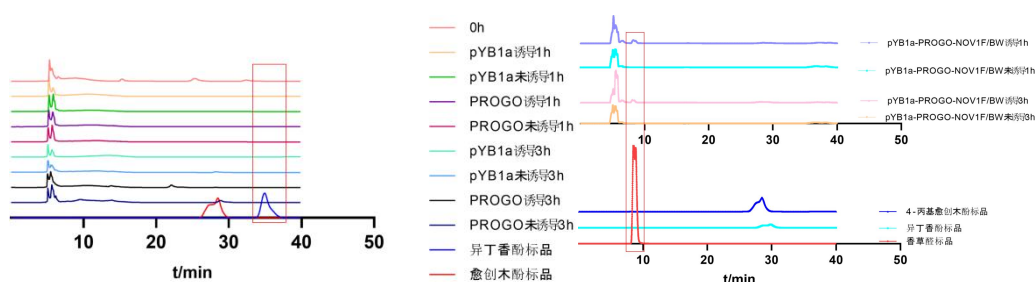
The mobile phase was prepared: methanol and 0.1% glacial acetic acid were also prepared.

2.Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

3.HPLC detection.

The samples were detected by HPLC using the methanol and 0.1% glacial acetic acid method. Column: Aminex XP-C18; flow rate: 0.5 mL/min; injection volume: 10 μ L.



Module 2:

(A)Experiment result 1 Summary — Whole-cell catalysis and HPLC detection of wild-type ODM

Take BW, pYB1a-*eGFP*, pYB1a-*odm* bacterial strains cultured at 37 °C for 12 h. Set non-ara-induction control for each group. Transfer six tubes of ZY medium into biosafety cabinet.

Prepare induction system: 50×M 100 μL, 5052 100 μL, MgSO₄ 10 μL, 1000× stock 10 μL, antibiotic 5 μL, bacterial culture 50 μL, 20% arabinose inducer 50 μL.

Induce at 30 °C in shaker for 16 h.

2. Whole-cell catalysis

Measure OD₆₀₀ by spectrophotometer, normalize cell density to OD₆₀₀=6. Formula: sampling volume (mL)=6 /measured OD₆₀₀. Aliquot bacterial culture into 1.5 mL centrifuge tubes according to calculated volume.

Balance centrifuge tubes strictly; centrifuge at 4200 rpm for 10 min; discard all supernatant after centrifugation.

Prepare 200 μ L reaction system: final vanillic acid (VA) concentration 2.5 mM; solvent is 50 mM Tris-HCl buffer pH 7.5. Add all reagents to cell-pellet tubes.

Set 0-h control sample for each group and store at -20°C . Incubate catalysis groups in shaker: 30°C , 200 rpm, dark, 12 h.

3. Sample preparation

Take out all post-catalysis samples; centrifuge 13000rpm for 10 min to fully pellet cells.

Carefully collect clear supernatant.

Filter supernatant through 0.22 μ m aqueous- phase filter membrane, collect filtrate into sample vials.

Transfer filtrate into new LC insert liners, label samples, place into reusable HPLC sample vials.

Store all samples at -20°C refrigerator before HPLC injection.

4. Mobile-phase preparation

Prepare mobile phase: 0.1% phosphoric acid aqueous solution : methanol =75:25, mix thoroughly. Also prepare 10% methanol and pure methanol, transfer into large bottles and label name & date.

This is aqueous-organic mixed phase; perform vacuum filtration for aqueous component into HPLC solvent reservoir to remove fine impurities.

After filtration, sonicate mobile phase for 30 min for degassing to eliminate dissolved bubbles, preventing baseline drift and ghost peaks in HPLC run.

5.HPLC parameters & primary result

Mobile phase: 0.1% aqueous phosphoric acid : methanol = 75: 25

Column: C18, 250 mm $\times$ 4.6 mm $\times$ 5 μ m

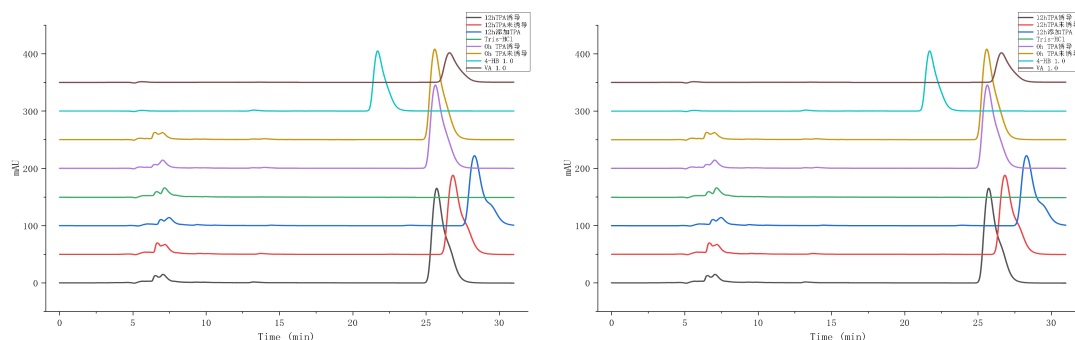
Detection wavelength: 254 nm

Injection volume: 5 μ L

Flow rate: 0.5 mL/min

Runtime per sample: 35 min

Catalysis solution: final VA concentration 2.5 mM, solvent 50 mM Tris-HCl pH 7.5



Neither induced nor non-induced ODM groups showed target peak of 4-HB (retention time ~21 min). Peak retention time shifted in induced-ODM group spiked with 4-HB control.

6.HPLC Improvements

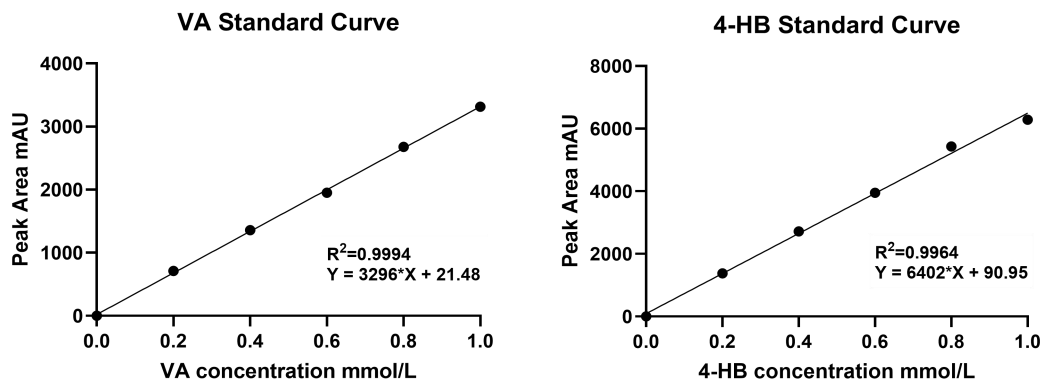
Adjust catalytic conditions: set 6 h and 18 h reaction groups. Each group contains ODM-induced, ODM-uninduced and blank transformation controls for condition exploration.

For the 12 h group, in addition to induced, uninduced and 4-HB-spiked groups, an extra control is supplemented: catalysis using 7 OD bacterial suspension after induction.

The 0 h catalytic group is performed as planned.

HPLC parameters are kept constant.

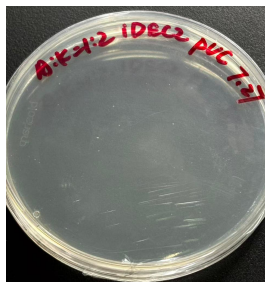
7.HPLC parameters & result with standard curves



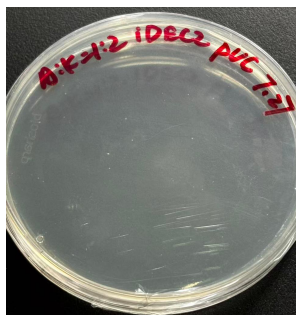
(B)Preparation and Efficiency Verification of *gYB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr*

Competent Cells

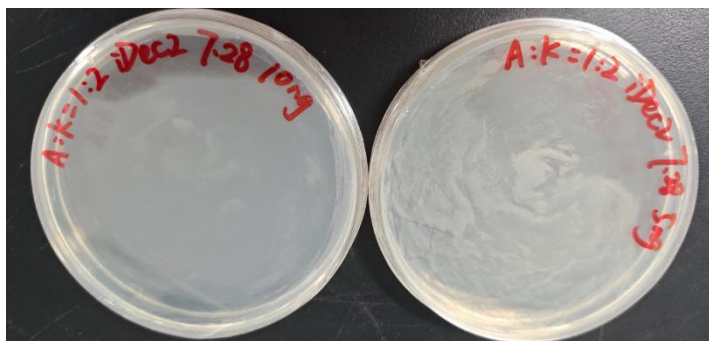
Competent cells were prepared by culturing the bacterial suspension at 30 °C for 1 h, then shifting to 16 °C for 11 h. Efficiency verification and concentration-gradient test results are shown below:



- Transformation with 1 pg pUC control plasmid (efficiency check)

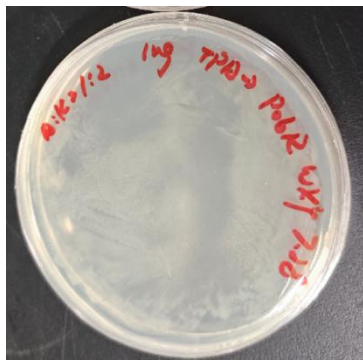


- Transformation with ddH₂O (contamination check)

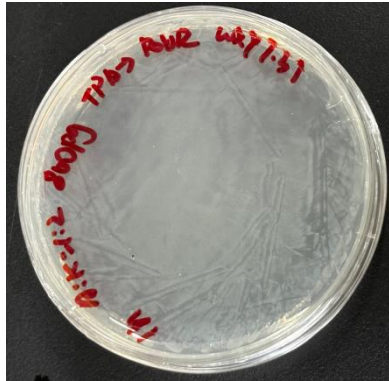


- Transformation with 10 ng ODM mixed plasmid

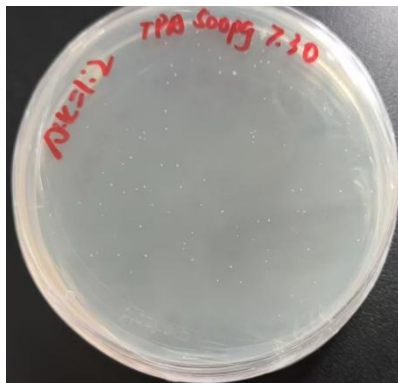
- Transformation with 5 ng ODM mixed plasmid



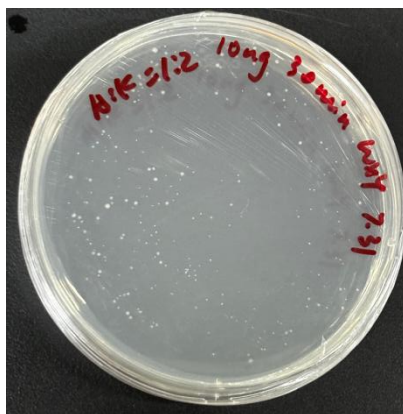
- Transformation with 1 ng ODM mixed plasmid



- Transformation with 10 ng ODM mixed plasmid, 30 min outgrowth: ~200 colonies

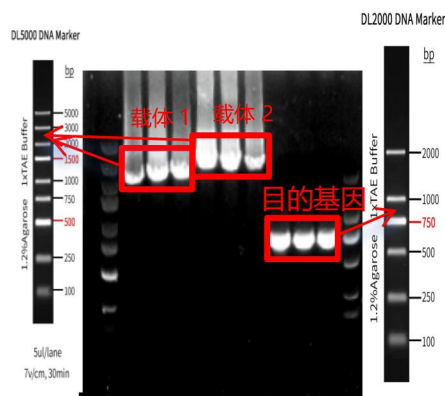


- Transformation with 500 pg ODM mixed plasmid: ~100 colonies



- Transformation with 800 pg ODM mixed plasmid

*gYB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr* and *pYB1a-odm* mixed plasmid were co-transformed into DH5 α and BW Δ CD. Results are as follows.



PCR was successful; products were purified and digested.

Reagent	Volume
Dpn I enzyme	0.2μL
CutSmart Buffer	1μL
Purified product	200 ng
Total volume	10μL

Conditions: 37 ° C for 2 h; 80 ° C for 20 min.

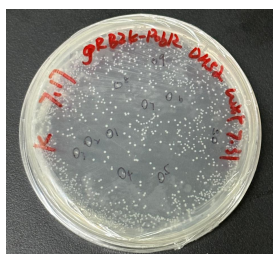
3.Gibson Assembly System

Assembly system (10 μL total):

4.Plating Results

Component	Volume (μL)
C117 enzyme mix	5
Target gene	1
Template fragment 1	2
Template fragment 2	2

Temperature	Time
50 °C	30 min
4 ° C	∞



-Recombinant plasmid transformed into DH5α competent cells

-Recombinant plasmid transformed into BWΔCD competent cells

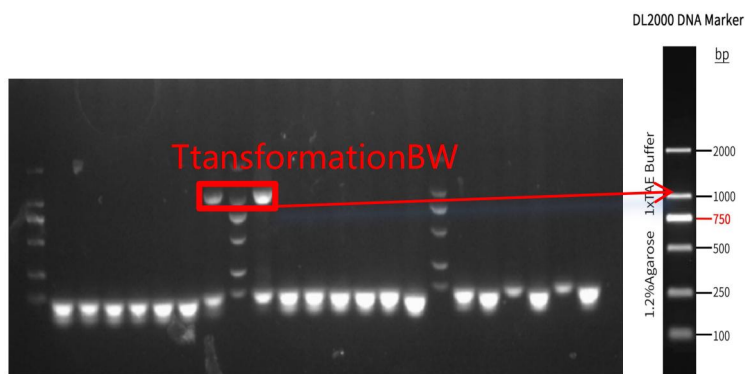
Ten colonies were picked from each plate for colony PCR.

5.Colony PCR System and Program

Colony PCR system (10 μL)	
Reagent	Volume (μL)
2× Green Mix	5
Forward primer (F)	0.4
Reverse primer (R)	0.4
ddH ₂ O	4.2

Temperature	Time	Cycles
98 °C	5 min	1
98 °C	30 s	30×
58 °C	30 s	30×
72 °C	32 s	30×
72 °C	5 min	1
4 °C	∞	—

6.Colony PCR Results (Ligation Considered Successful) and Plasmid Extraction



Target length: 1050 bp.

Colony PCR bands were correct; ligation is considered successful.

Colony 7:327.4 ng/ μ L ; Colony 8:369.2 ng/ μ L

Colonies 7 and 8 were transformed into BW Δ CD.

Module 3:

(A) HPLC validation of the vanillic acid-to-pyruvate pathway

1. Induction

BW, BW Δ gldA Δ maeB Δ pykF-pLB1s-VanAB-aroYBW Δ gldA Δ maeB Δ pykF-pYB1a-dmpB-dmpD-mhpD-mhpE(full pathway), BW Δ gldA Δ maeB Δ pykF-pYB1a-dmpB-mhpD-mhpE(downstream pathway) and BW Δ gldA Δ maeB Δ pykF-pYB1a-eGFP (empty vector) cells cultivated at 37°C for 12 hours were taken out; a control group without arabinose (ara) induction was set for each strain, and eight tubes of ZY medium were taken into the clean bench. The induction system was configured according to the following ratios: ZY induction system:

Component	Volume
50 $\times$ M	100 μ L
5052	100 μ L
MgSO ₄	10 μ L
1000 $\times$	10 μ L
Antibiotics	5 μ L
Bacterial fluid	50 μ L
20% ara	50 μ L

The cultures were induced in a shaking incubator at 30°C for 16 hours.

2. Whole-cell catalysis

The OD value was measured with a spectrophotometer, and the cell concentration was uniformly calibrated to OD=6; sampling volume (mL) = 6 $\div$ measured OD value. The bacterial fluid was dispensed into 1.5 mL centrifuge tubes according to the calculated volume. The tubes were strictly balanced and centrifuged at 4200 rpm for 10 min; after centrifugation, the supernatant was completely discarded. The reaction systems were configured as follows:

(1) Full pathway BW Δ gldA Δ maeB Δ pykF-pLB1s-VanAB-aroY-dmpD BW Δ gldA Δ maeB Δ pykF-pYB1a-dmpB--dmpD-mhpD-mhpE: final concentration of 5mM vanillic acid, final concentration of 10mM NADH, with 5mM H₂SO₄ solution as the buffer. All components were added to the cell-pellet tube and mixed by pipetting.

(2) Downstream pathway BW Δ gldA Δ maeB Δ pykF-pYB1a-dmpB-dmpD-mhpD-mhpE final concentration of 5mM catechol, final concentration of 0.5mM ferrous ion, final concentration of 1mM manganese ion, with 5mM H₂SO₄ solution as the buffer. All components were added to the cell-pellet tube and mixed by pipetting. The tubes were placed in a shaking incubator at 30°C, 200 r/min, in the dark, for 4h and 6h.

3. Sample preparation

All catalysis-finished samples were taken out and centrifuged at 4200 rpm for 10 min to completely precipitate the cells. The clear reaction supernatant was carefully aspirated. The supernatant was filtered through a 0.22 μ m aqueous membrane to remove impurities and collected into collection vials. The filtrate was loaded into new disposable HPLC insert tubes, the samples were labelled, and placed into reusable HPLC vials. All samples were stored at -20°C for later HPLC analysis.

4. Mobile phase configuration and HPLC method

Mobile phase configuration:

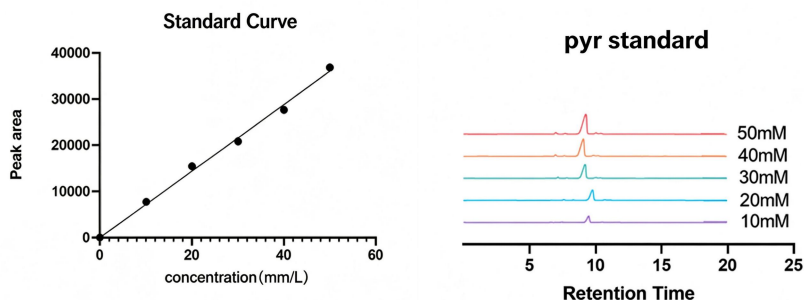
(1) A 5mM H₂SO₄ mobile phase was configured, filtered through an aqueous membrane, and sonicated for 30 min.

(2) 5% acetonitrile/5mM H₂SO₄ and 30% acetonitrile/5mM H₂SO₄ were configured, filtered through an aqueous membrane, and sonicated for 30 min.

HPLC method: mobile phase, 5mM H₂SO₄ aqueous solution; flow rate, 0.6 mL/min; column temperature, 50°C; column type, Aminex HPX-87H; injection volume, 10μL.

5. HPLC results

5.1 Pyruvate standard curve



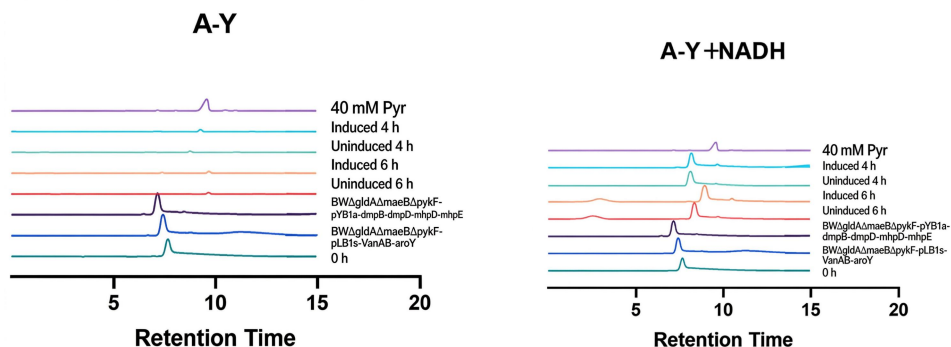
$$Y=720.2X, R^2\approx 0.9949.$$

5.2 Full-pathway validation

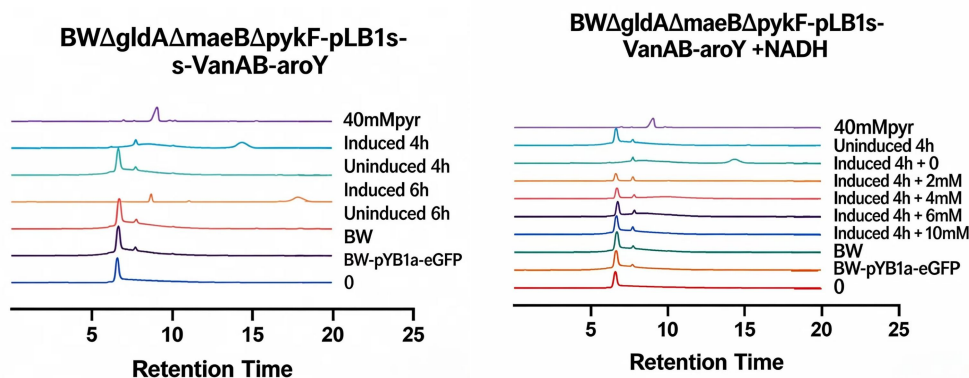
A-Y: pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE/BW; empty vector: pYB1a-eGFP/BW; √: induced with ara, ×: without ara; 4: catalysis 4h, 6: catalysis 6h, 0: 0h transformation solution.

Round 1 and Round 2.

(1) In round 1, the substrate VA of the group without NADH was completely consumed; by contrast, the full-pathway substrate VA of the group with NADH was not obviously consumed.



(2) In round 2, an NADH gradient was set; with increasing NADH, the substrate VA consumption decreased, and at 10mM NADH there was almost no consumption.

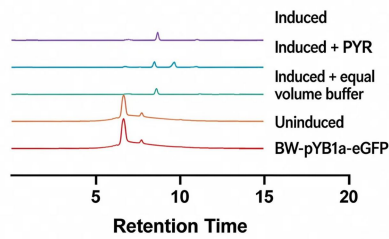


(3) The substrate-consumption problem of the uninduced group in round 1 was effectively solved in round 2; the induced and uninduced groups were clearly distinguished.

5.3 Full-pathway product validation

In the induced, 4h whole-cell catalysis parallel system of A-Y, the following treatments were performed: no additional addition, addition of 10mM PYR, and addition of an equal volume of buffer. As shown in the figure, the substrate VA was obviously consumed, but no PYR was produced.

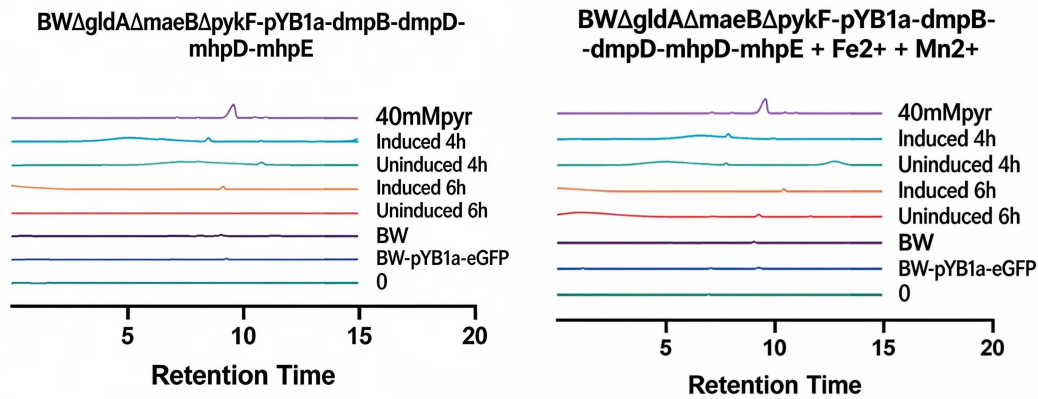
Pyruvate production validation



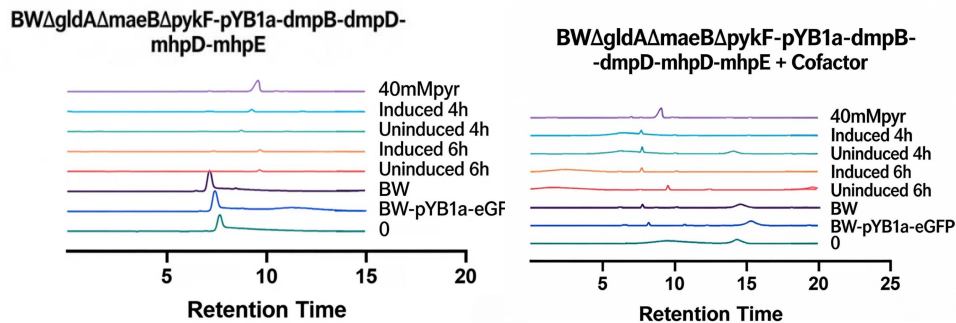
5.4 Downstream pathway validation

B-E: pYB1a-dmpBD-mhpDE/BW; empty vector: pYB1a-eGFP/BW.

(1) In round 1, the substrate CA had no obvious peak and could not be detected.



(2) In round 2, a non-sharp peak at about 15 min might be the substrate peak, but the induced-4h-without-cofactor group and the uninduced-4h-with-cofactor group both had similar peaks; the 6h groups (induced/uninduced, with/without cofactors) all had no similar peak.



(3) It was judged that no PYR was produced.

(B) Characterization and verification using the pyruvate sensor

1. 1×M9 medium preparation

1.1 Preparation of neutral vanillic acid solution

5.8 mL of 10mM vanillic acid stock solution and 34.2 mL of ddH₂O were aspirated and mixed to prepare 40 mL of solution; 1 mol/L NaOH was slowly added dropwise to adjust the pH to 7.02.

1.2 Preparation of antibiotic-free 1×M9 base solution (150 mL)

Antibiotic-free 1×M9 base solution (150 mL):

Component	Volume
5×M9 salt solution	24 mL
20% glycerol	15 mL
1M MgSO ₄	300μL
Sterile ddH ₂ O	109.7 mL
1M CaCl ₂	15μL

24 mL of 5×M9 salt solution, 15 mL of 20% glycerol, 300μL of 1M MgSO₄ and 109.7 mL of sterile ddH₂O were added; 15μL of 1M CaCl₂ was slowly added and mixed gently.

1.3 Preparation of 1×M9 medium containing vanillic acid (50 mL)

34.48 mL of neutral vanillic acid solution was aspirated, and 15.52 mL of sterile antibiotic-free 1×M9 was added to the flask; the contents were gently inverted to mix; the final vanillic acid

concentration was 1mM.

2. Bacterial fluid treatment

The BW original strain and BW- Δ gldA- Δ maeB- Δ pykF (pyruvate sensor) that did not need induction were shaken in LB medium for 16h; the pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE that needed induction was shaken and then induced in ZY medium for 16h. Bacterial fluid with OD=6 was taken into 1.5 mL EP tubes and centrifuged at 4200 rpm for 10 min; the supernatant was removed; 600 μ L of antibiotic-free 1 $\times$ M9 medium was added and mixed by pipetting; the tubes were centrifuged at 4200 rpm for 10 min; the supernatant was removed; 600 μ L of antibiotic-free 1 $\times$ M9 medium was added and mixed by pipetting.

3. Addition system

A total of 11 groups were set up, as shown in the following table:

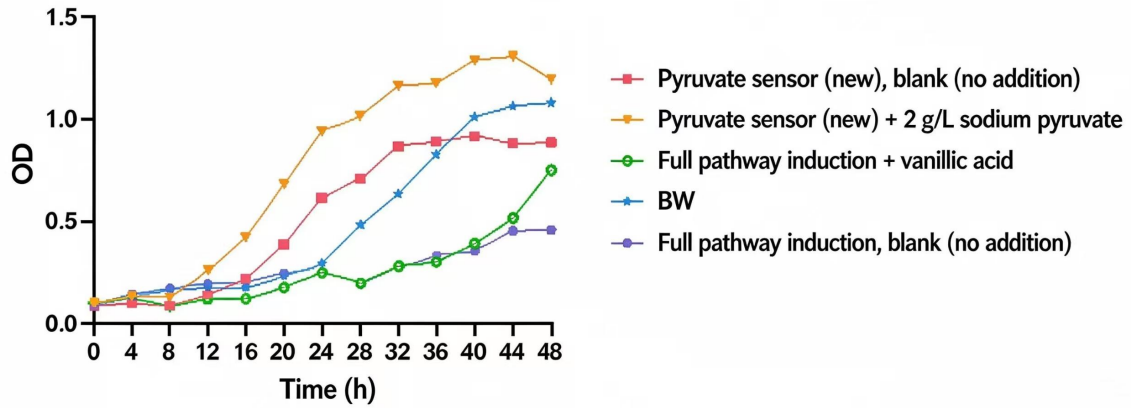
Group	1 $\times$ M9 medium	Additives	Bacterial fluid
BW	4950 μ L ordinary antibiotic-free 1 $\times$ M9	—	50 μ L BW fluid
Pyruvate sensor (new)	4950 μ L ordinary antibiotic-free 1 $\times$ M9	—	50 μ L pyruvate sensor fluid (new)
Pyruvate sensor (old)	4950 μ L ordinary antibiotic-free 1 $\times$ M9	—	50 μ L pyruvate sensor fluid (old)
Pyruvate sensor (new) + 2 g/L sodium pyruvate	3950 μ L ordinary antibiotic-free 1 $\times$ M9	1000 μ L 10 g/L sodium pyruvate stock	50 μ L pyruvate sensor fluid (new)
5Pyruvate sensor (old) + 2 g/L sodium pyruvate	3950 μ L ordinary antibiotic-free 1 $\times$ M9	1000 μ L 10 g/L sodium pyruvate stock	50 μ L pyruvate sensor fluid (old)
Full pathway (induced) + vanillic acid (AS)	4950 μ L 1 $\times$ M9 containing vanillic acid	5 μ L AS antibiotic	50 μ L full-pathway fluid
Full pathway (induced), no vanillic acid (AS)	4950 μ L ordinary antibiotic-free 1 $\times$ M9	5 μ L AS antibiotic	50 μ L full-pathway fluid
Pyruvate sensor (new), 1 g/L sodium pyruvate	4500 μ L antibiotic-free 1 $\times$ M9	500 μ L 10 g/L sodium pyruvate stock	50 μ L sensor fluid (new)
Pyruvate sensor (old), 1 g/L sodium pyruvate	4500 μ L antibiotic-free 1 $\times$ M9	500 μ L 10 g/L sodium pyruvate stock	50 μ L sensor fluid (old)
Pyruvate sensor (new), 3 g/L sodium pyruvate	3500 μ L antibiotic-free 1 $\times$ M9	1500 μ L 10 g/L sodium pyruvate stock	50 μ L sensor fluid (new)
Pyruvate sensor (old), 3 g/L sodium pyruvate	3500 μ L antibiotic-free 1 $\times$ M9	1500 μ L 10 g/L sodium pyruvate stock	50 μ L sensor fluid (old)

4. Results and analysis

4.1 Full-pathway verification using the pyruvate sensor (induced group)

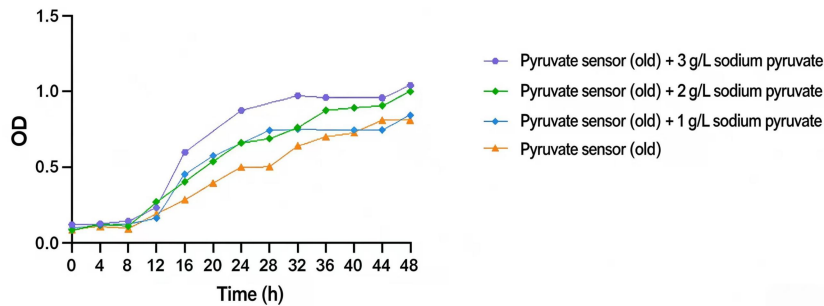
For the pyruvate sensor carrying the pLB1s-VanAB-aroY and pYB1a-dmpB-dmpD-mhpD-mhpE plasmids, after induction, the sensor strain with exogenous 2 g/L sodium pyruvate grew best, indicating that the sensor strain can effectively respond to the pyruvate substrate. BW grew slower initially but eventually better than the blank-group pyruvate sensor strain. After the vanillic acid substrate was applied, the full-pathway induced strain grew faster in the second half than the blank-group full-pathway induced strain (protein expression and enzymatic reactions in the first half may create growth pressure).

BWΔgldAΔmaeBΔpykF-pLB1s-VanAB-aroY
 BWΔgldAΔmaeBΔpykF-pYB1a-dmpB-dmpD-mhpD-mhpE
 Post-induction

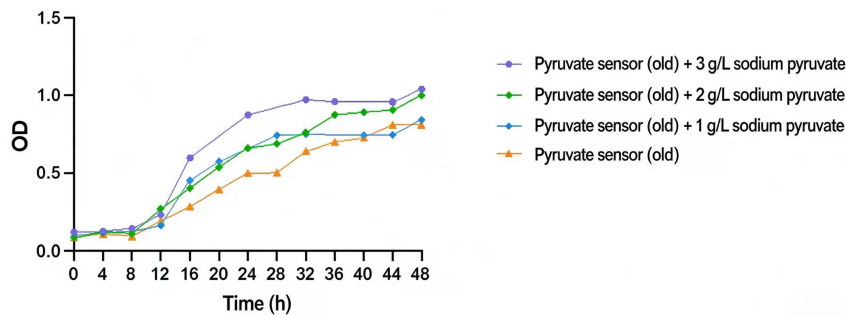


4.2 Old pyruvate sensor characterization (cultured and monitored in M9 medium)

Growth curves of bw-ΔgldA-ΔmaeB-ΔpykF at different sodium pyruvate concentrations.

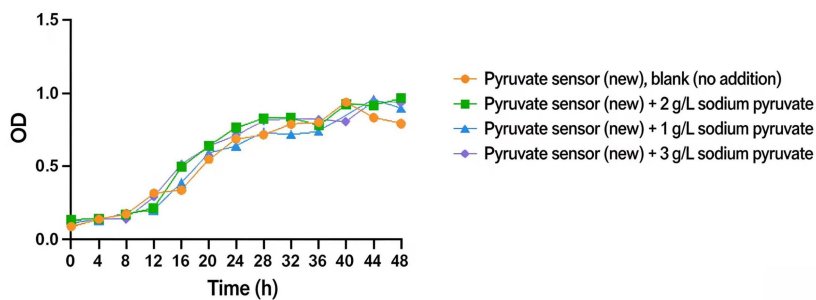


Round 1: 0 g/L < 1 g/L < 2 g/L < 3 g/L;

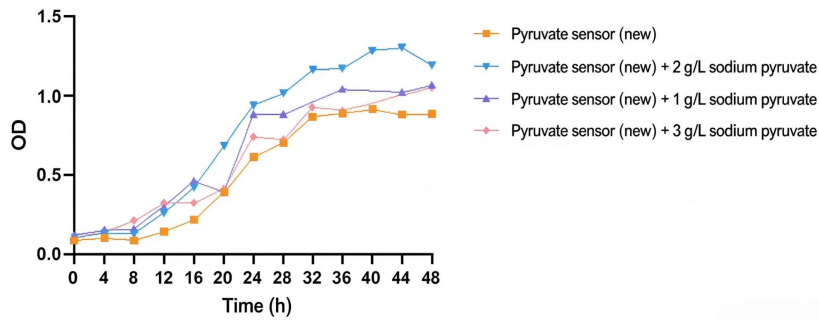


Round 2: 0 g/L < 1 g/L < 2 g/L < 3 g/L. The two rounds are consistent.

4.3 New pyruvate sensor characterization



Round 1: 0 g/L < 3 g/L < 1 g/L < 2 g/L;



Round 2: the effect was not good.

Week 11 (8.3-8.6)

Module 1:

1. Inoculation

Original strains : BW-pYB1a-eGFP 、 BW-pYB1a-progo-nov1f 、 BW-pYB1a-nov1f 、 BW-pYB1a-progo

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-eGFP、 BW-pYB1a-progo-nov1f、 BW-pYB1a-nov1f、 BW-pYB1a-progo

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

Using 4-PG to catalyze the induced enzymes for Isoeugenol production and Isoeugenol to catalyze the induced enzymes for vanillin production, vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Tris-HCl(pH=7.5), and transfer to a centrifuge tube for whole-cell

catalysis.

Incubate in a shaker at 30°C for 1h, 3h hours.

4.Prepare mobile phase, standards and filter samples

Preparation of the mobile phase.

The mobile phase was prepared: Methanol: 0.1% acetic acid (ice) = 70:30.

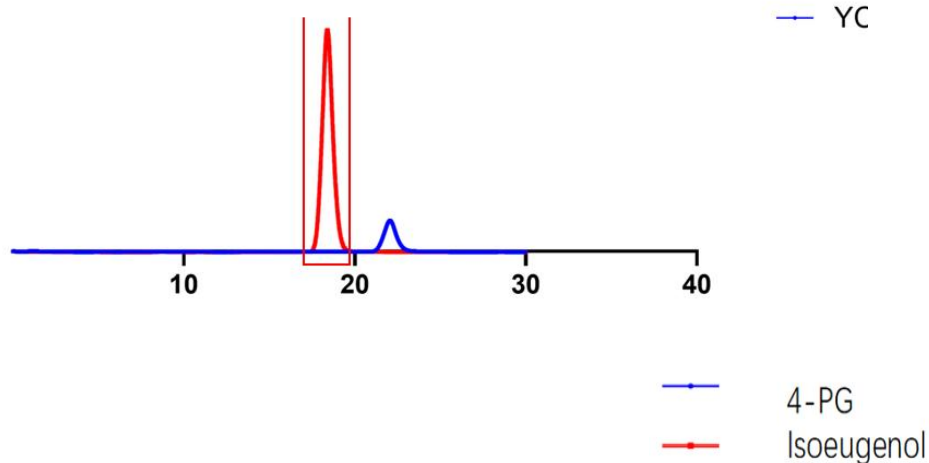
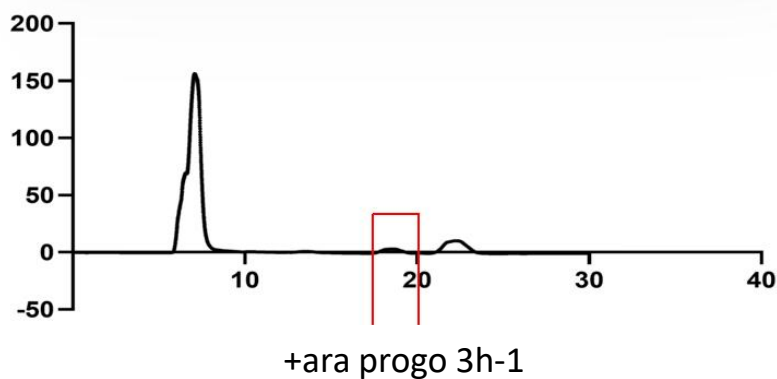
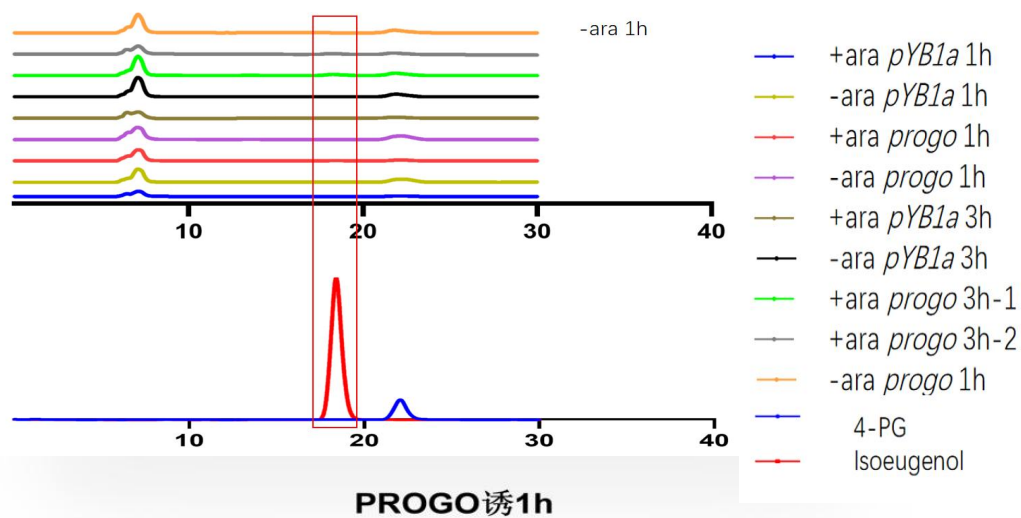
Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22μm membranes for HPLC.

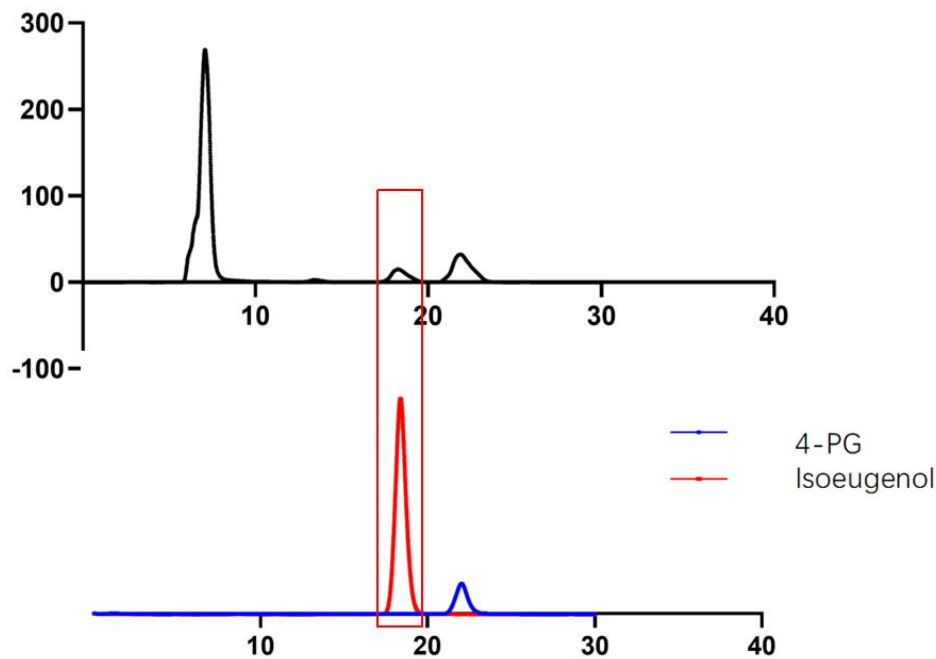
5.HPLC run (sulfuric acid method / phosphoric acid method)

The samples were detected by HPLC using the sulfuric acid method (mobile phase: 5mM H₂SO₄) and the phosphoric acid method. Column: XP-C18; flow rate: 0.4 mL/min; injection volume: 10μL.

Detection of Isoeugenol



+ara progo 1h



Week2:8.7-8.13

(A) Verification of Expression Efficiency

1. Inoculation

Original strains: BW-pYB1a-eGFP, BW-pYB1a-, BW-pYB1a-nov1f, BW-pYB1a-progo

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-eGFP, BW-pYB1a-progo-nov1f, BW-pYB1a-nov1f, BW-pYB1a-progo

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

Using 4-PG to catalyze the induced enzymes for Isoeugenol production and Isoeugenol to catalyze the induced enzymes for vanillin production, vanillin to catalyze the induced enzymes for VA

production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Potassium phosphate buffer solution(PH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

Incubate in a shaker at 30°C for 1h, 3h, 6h, 9h hours.

4.Prepare mobile phase, standards and filter samples

Preparation of the mobile phase.

The mobile phase was prepared: Methanol: 0.1% acetic acid (ice) = 70:30. Methanol: 0.1% acetic acid (ice) = 60:40. Methanol: 0.1% acetic acid (ice) = 50:50.

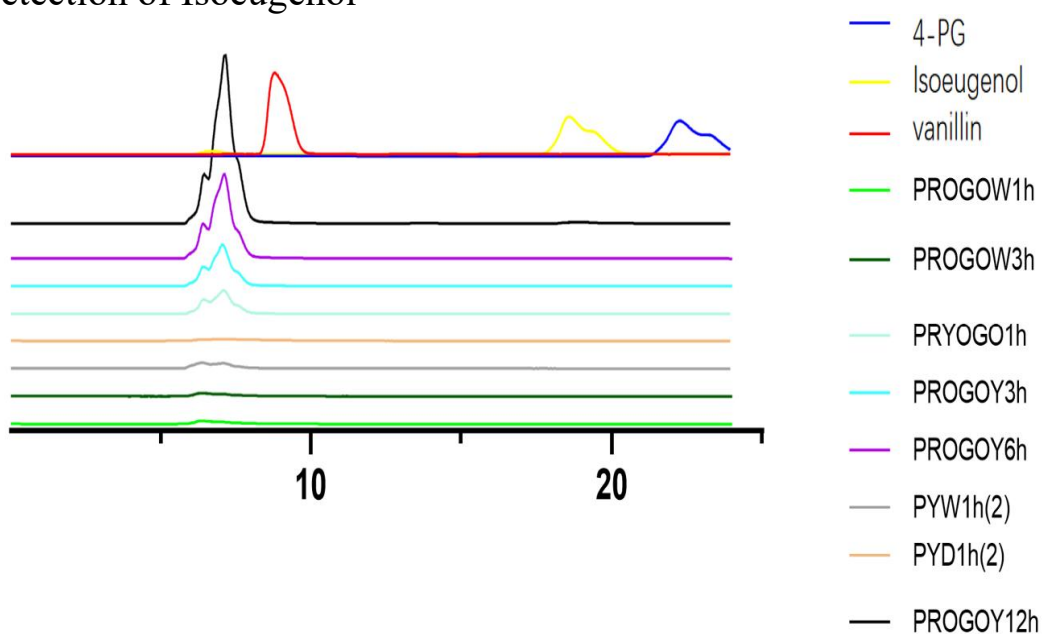
Preparation of standards and sample filtration.

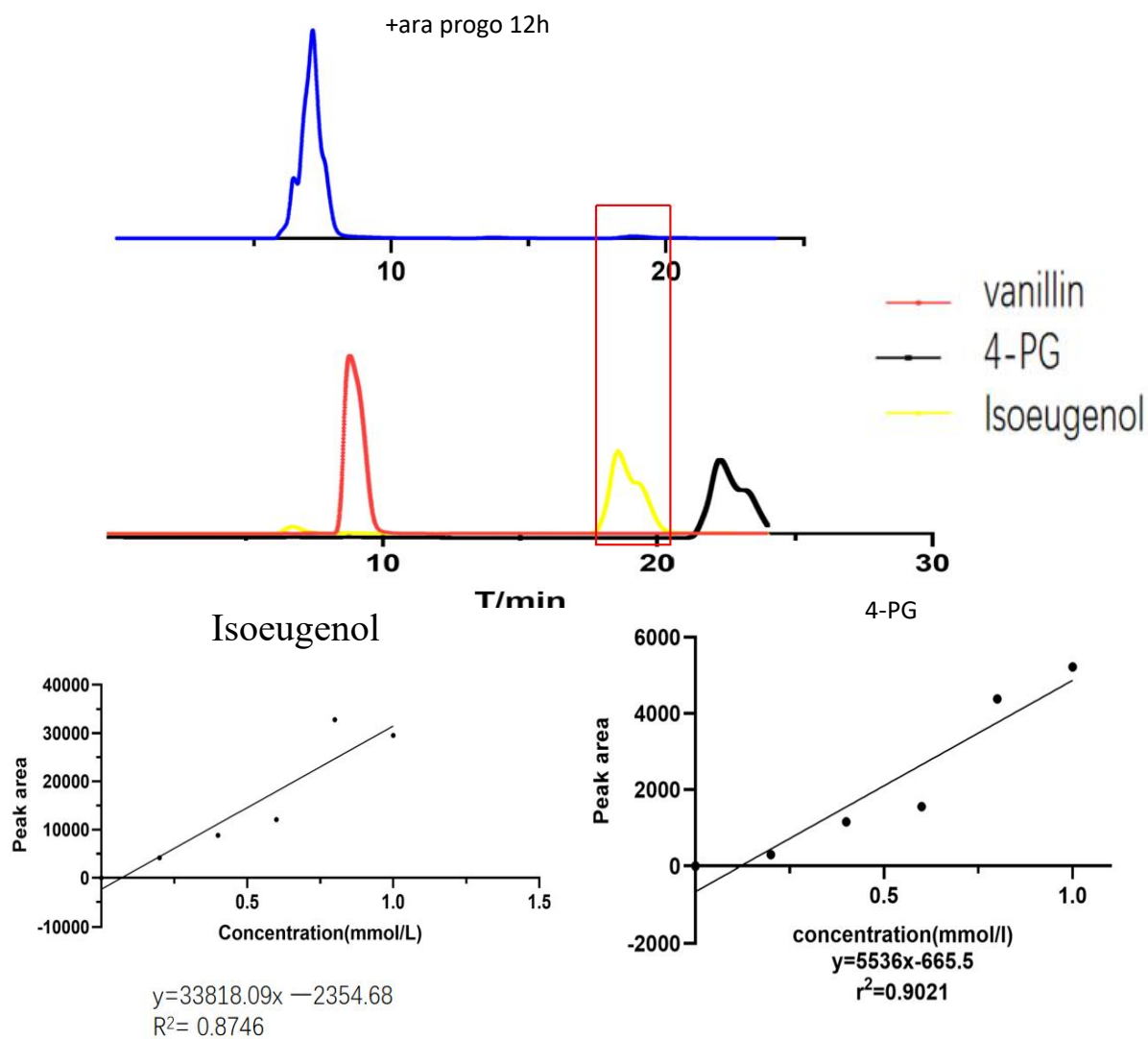
The standard solutions were prepared, and the samples were filtered through 0.22μm membranes for HPLC.

5.HPLC run (sulfuric acid method / phosphoric acid method)

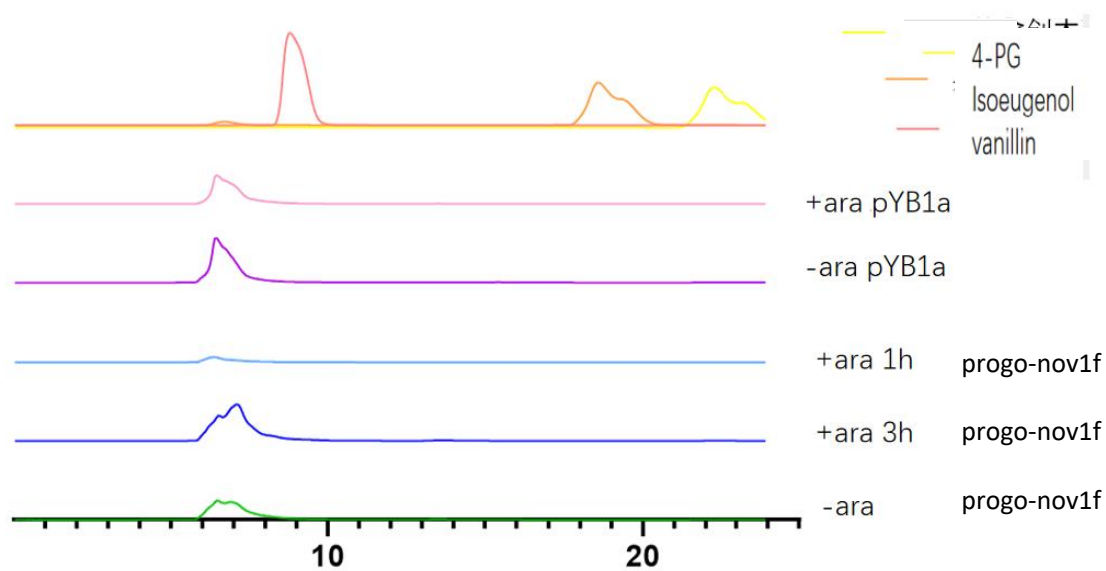
The samples were detected by HPLC using the sulfuric acid method (mobile phase: 5mM H₂SO₄) and the phosphoric acid method. Column: XP-C18; flow rate: 0.4 mL/min、0.5 mL/min; injection volume: 10μL.

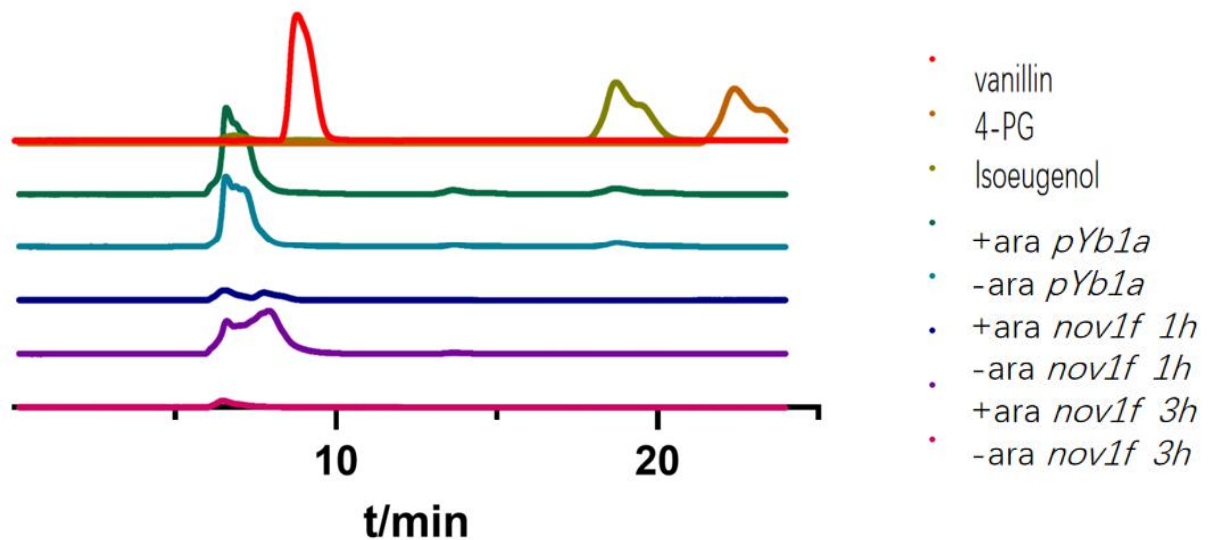
Detection of Isoeugenol





Detection of vanillin



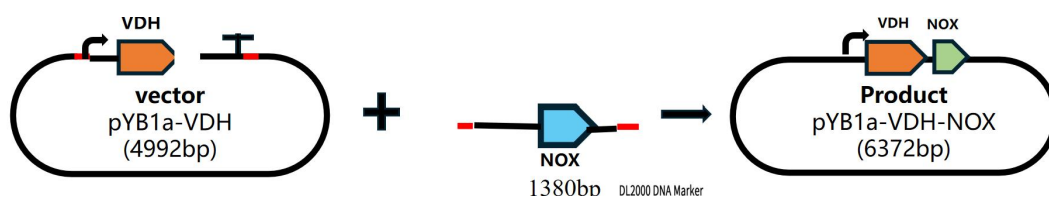


(B) Construct BW-pYB1a-*vdh-nox*

1. PCR + cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and enzymatic digestion of vector *pYB1a*

Subsequently, the relevant DNA fragments were assembled using Enzymatic cleavage ligation, followed by transformation into competent cells. Upon completion of the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed.

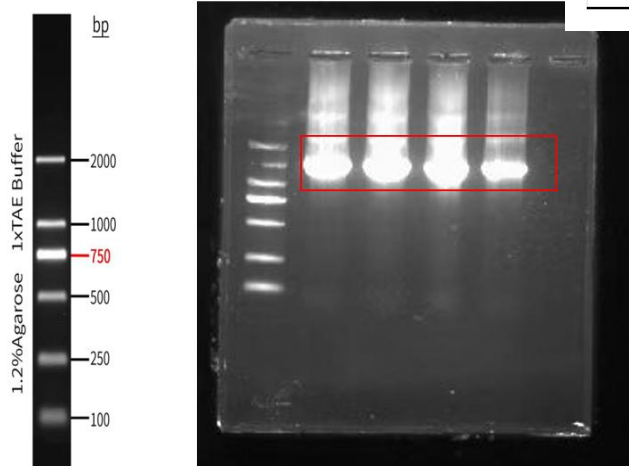


PCR system (50μL)	
2×Mix	25ul
<i>nox</i> -F	2ul
<i>nox</i> -R	2ul
Template	20ng
DDW	To 50ul

PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	41s
72°C	5min
25°C	∞

} × 30

DL2000 DNA Marker

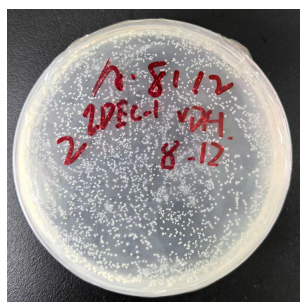


2. Connection

Connecting the two fragments using the Enzymatic cleavage ligation.

3. Transformation

The ligated system was transferred into DH5 α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



Module 2:

(A) Whole-Cell Catalysis and HPLC Sample Preparation of Wild-Type ODM Under Optimized Conditions

1. Induction

BW, pYB1a-*eGFP*, and pYB1a-*odm* cultures grown at 37 °C for 12 h were taken out. For each group, a non-arabinose-induced control was set. Six tubes of ZY medium were brought into the laminar flow hood.

Induction system was prepared as follows: 50×M 100 μ L, 5052 100 μ L, MgSO₄ 10 μ L, 1000× stock 10 μ L, antibiotic 5 μ L, bacterial culture 50 μ L, 20% arabinose inducer 50 μ L.

Induced in a 30 °C shaker for 16 h.

2. Whole-cell catalysis

OD₆₀₀ was measured with a spectrophotometer and normalized to OD = 6. Calculation: sampling volume (mL) = 6 $\div$ measured OD. The bacterial suspension was aliquoted into 1.5 mL centrifuge tubes accordingly.

Tubes were carefully balanced and centrifuged at 4200 rpm for 10 min; all supernatant was discarded.

A 200 μ L reaction system was prepared: vanillic acid (VA) at a final concentration of 2.5 mM in 50 mM Tris-HCl buffer (pH 7.5). The solution was added to each cell-pellet tube.

A 0 h control for each group was stored at -20 °C. Catalysis groups were incubated in a shaker at 30 °C, 200 r/min, protected from light, for 12 h.

3. Sample preparation

All post-catalysis samples were taken out and centrifuged at 13,000 rpm for 10 min to fully pellet the cells.

The clear supernatant was carefully pipetted.

The supernatant was filtered through a 0.22 μ m aqueous filter membrane to remove impurities and collected in vials.

The filtrate was transferred into new disposable HPLC inserts, properly labeled, and placed into reusable HPLC sample vials.

All samples were stored at -20 °C pending HPLC injection.

4. Mobile phase preparation

Mobile phase was prepared at the ratio: 0.1% phosphoric acid aqueous solution : methanol = 75:25, mixed thoroughly. 10% methanol and pure methanol were also prepared, aliquoted into large bottles labeled with name and date.

This is an aqueous-organic mixed phase; the aqueous component was vacuum-filtered into the HPLC solvent reservoir to remove fine particles.

After filtration, the mobile phase was sonicated for 30 min for degassing to completely remove dissolved bubbles, preventing baseline instability and ghost peaks during HPLC runs.

5. Setup and Preparation Work

Sample preparation conditions:

Catalytic conditions were modified: 6 h and 18 h catalysis groups were set, each containing OMD-induced, ODM-uninduced, and empty-vector control subgroups.

In the 12 h group, beyond the normal induced, uninduced, and 4-HB-spiked subgroups, an additional control was added: catalysis with 8 OD bacterial suspension after induction.

The 0 h catalysis group was performed as usual.

HPLC conditions were confirmed unchanged.

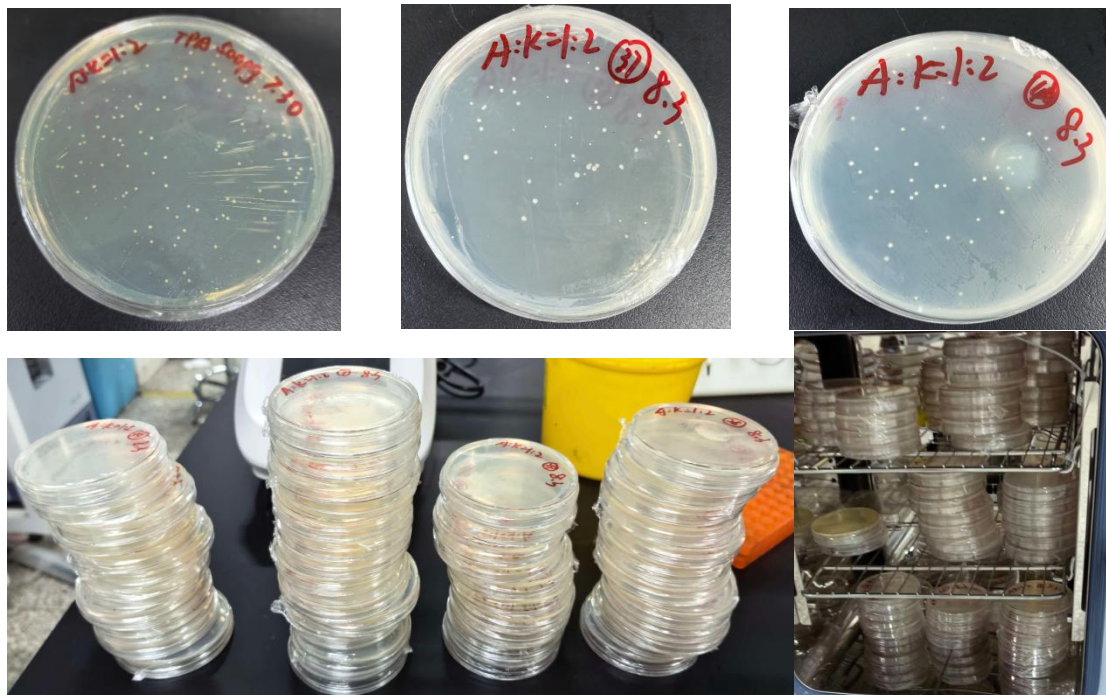
(B) Library Construction and Biosensor-Based High-Throughput Screening of pYB1a-*odm* Mutants

1. Large-Scale Transformation of pYB1a-*odm* Mixed Plasmid into gYB2k-*pobR*^{WT}-P_{pobA}*2-*mCherry*-CD-Cmr Competent Cells

Mixed plasmid concentration: 10 ng/ μ L. Competent cells were incubated on ice for 5 min, then 1 μ L mixed plasmid was added.

Incubated on ice for 20 min, heat-shocked at 42 °C for 45 s, then immediately placed on ice for 2–3 min.

After incubation, 600 μL LB was added to each EP tube and shaken at 37 $^{\circ}\text{C}$ for 45 min.
Inside the biosafety cabinet, 100 μL was pipetted, mixed, and spread onto plates.



100 plates were co-transformed, with ~ 100 colonies per plate.

Library capacity: 1×10^5 clones.

2. Screening Assay for *pYB1a-odm* / *gYB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr*

M9 selection system was set up. Bacterial culture, vanillic acid, and chloramphenicol concentrations were tested:

- (1) VA concentration: 2.5 mM (10 mM stock solution, adjusted to pH ~ 7 with NaOH after dilution; diluted pH = 3.3)
- (2) Bacterial culture volume: 50 μL and 250 μL
- (3) Chloramphenicol concentration: 30 mg/mL

Detailed system (5 mL)

Component	WT group	<i>pobR</i> group	BW Δ CD group	Mixed (AK) group
Antibiotic	A 5 μL	K 5 μL	None	A 5 μL , K 5 μL
5 $\times$ M9	1 mL	1 mL	1 mL	1 mL
20% Glycerol	500 μL	500 μL	500 μL	500 μL
MgSO ₄	10 μL	10 μL	10 μL	10 μL
CaCl ₂	0.5 μL	0.5 μL	0.5 μL	0.5 μL
Chloramphenicol (C)	5 μL	5 μL	5 μL	5 μL
1. Water + VA + bacteria	3429.5 μL + 50 μL bacteria	3429.5 μL + 50 μL bacteria	3434.5 μL + 50 μL bacteria	3424.5 μL + 50 μL bacteria
2. Water + VA + bacteria	3229.5 μL + 250 μL bacteria	3229.5 μL + 250 μL bacteria	3234.5 μL + 250 μL bacteria	3224.5 μL + 250 μL bacteria

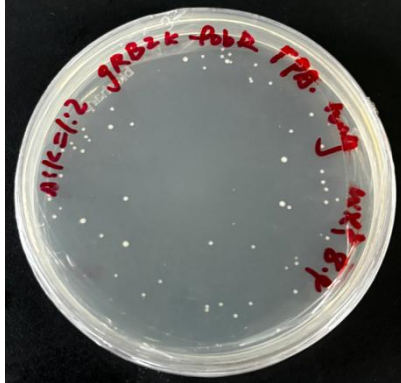
Question: When no obvious change is observed after subsequent subculturing, how much additional chloramphenicol should be added?

(C) New *gRB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr*

1. Sequencing Results of *gRB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr*; Ligation Successful



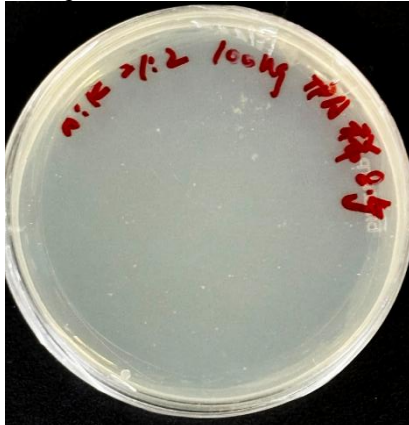
- Competent cells transformed with ddH₂O (contamination control)



- Competent cells transformed with 10 ng ODM



- Competent cells transformed with 100 ng ODM (diluted plate): ~30 colonies



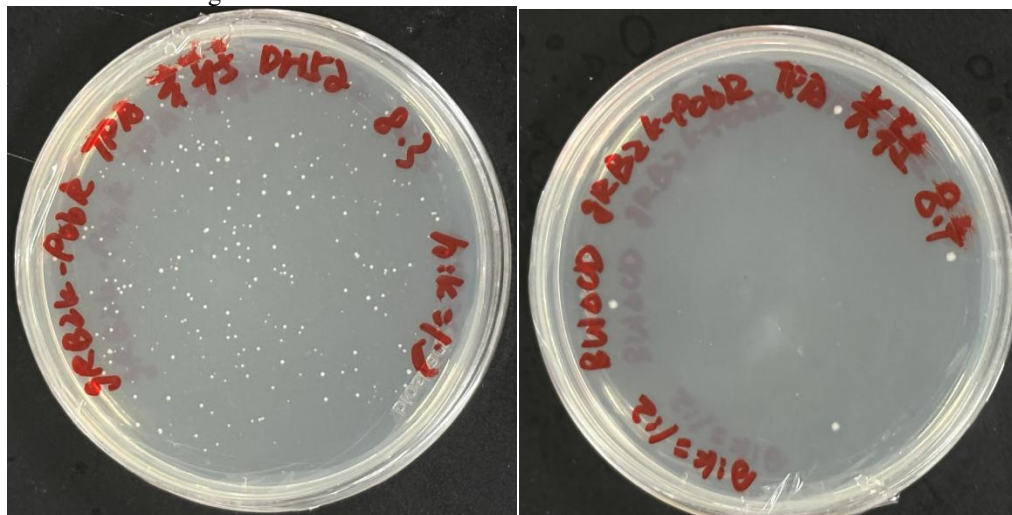
- Competent cells transformed with 100 ng ODM (concentrated plate): ~400 colonies



- Competent cells transformed with 50 ng ODM

3.Co-Transformation of gRB2k-pobR^{WT}-P_{pobA}*2-mCherry-CD-Cmr with pYB1a-odm into DH5α and BWΔCD

Results after 22 h of growth:



- Co-transformation into DH5α
- Co-transformation into BWΔCD

Module 3:

(A) HPLC sample preparation — induction and whole-cell catalysis

1. Induction.

The strains cultured at 37 °C for 12 h (BW, BW Δ gldA Δ maeB Δ pykF-pLB1s-VanAB-aroYBW Δ gldA Δ maeB Δ pykF-pYB1a-dmpB-dmpD-mhpD-mhpE, BW Δ gldA Δ maeB Δ pykF-pYB1a-dmpB-mhpD-mhpE) were taken out. A control without arabinose induction was set for each group, and eight tubes of ZY medium were placed into the clean bench. The induction system was configured as follows:

50×M	100 μL
5052 supplement	100 μL
MgSO ₄	10 μL
1000× trace element	10 μL
Antibiotic	5 μL
Bacterial suspension	50 μL
20% arabinose (inducer)	50 μL

and cultivated in a shaking incubator at 30 °C for 16 h.

2. Whole-cell catalysis.

The OD value was measured with a spectrophotometer, and the cell density was normalized to OD = 6. The volume of each aliquot was calculated as: sample volume (mL) = 6 ÷ measured OD, and the bacterial suspension was dispensed into 1.5 mL centrifuge tubes. After strict balancing, the tubes were centrifuged at 4200 rpm for 10 min, and all the supernatant was discarded. The reaction systems were then prepared as follows: (i) for the full-pathway strain (pLB1s-VanAB-aroY/pYB1a-dmpBD-mhpDE): 5 mM vanillic acid (final concentration), 10 mM NADH (final concentration), with 5 mM sulfuric acid solution as the solvent; (ii) for the downstream-pathway strain (pYB1a-dmpBD-mhpDE): 5 mM catechol (final concentration), 0.5 mM Fe²⁺ (final concentration), 1 mM Mn²⁺ (final concentration), with 5 mM sulfuric acid solution as the solvent. All the components were added to the cell pellet and mixed thoroughly by pipetting. The tubes were placed in a shaking incubator at 30 °C and 200 r/min in the dark for 1 h and 2 h.

(B) HPLC preparation — sample preparation, mobile phase and HPLC method

1. Sample preparation.

All the catalysis samples were taken out and centrifuged at 4200 rpm for 10 min to pellet the cells completely. The clear supernatant was carefully collected and filtered through a 0.22 μm aqueous membrane to remove impurities, and then collected into collection vials. The filtrate was loaded into disposable HPLC injection vials with inner tubes, marked, and placed into reusable HPLC vials. All samples were stored at -20 °C until analysis.

2. Mobile phase preparation.

The 5 mM sulfuric acid mobile phase was prepared, together with 5% acetonitrile in 5 mM sulfuric acid and 30% acetonitrile in 5 mM sulfuric acid. Each mobile phase was filtered through an

aqueous membrane and sonicated for 30 min.

3. HPLC method.

Mobile phase: 5 mM H₂SO₄; flow rate: 0.6 mL/min; column temperature: 50 °C; column: Aminex HPX-87H; injection volume: 10 µL.

(C) HPLC results — standards

1. Pyruvate and catechol (CA) standards.

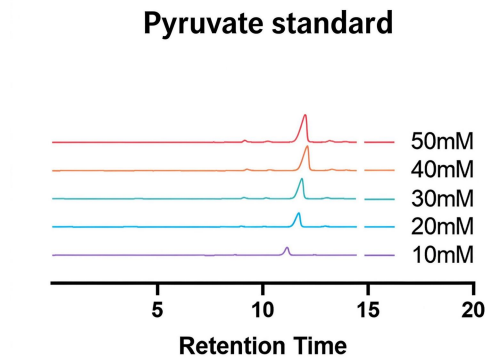


Figure 1 HPLC chromatograms of pyruvate standards (10–50 mM)

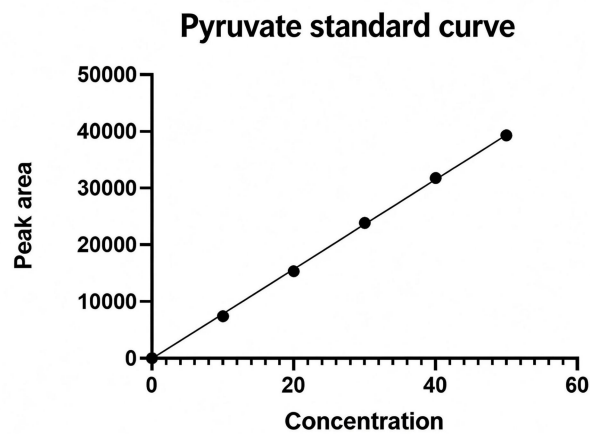


Figure 2 Standard curve of pyruvate ($Y = 788.4X$, $R^2 = 0.9996$)

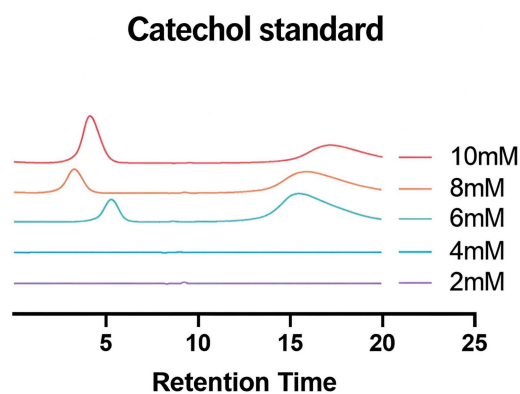


Figure 3 HPLC chromatograms of catechol (CA) standards

2. Acetic acid and lactic acid standards.

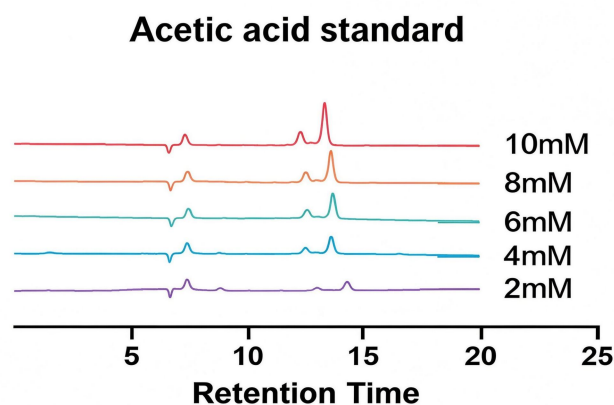


Figure 4 HPLC chromatograms and standard curve of acetic acid ($Y = 41.21X$, $R^2 = 0.9825$)

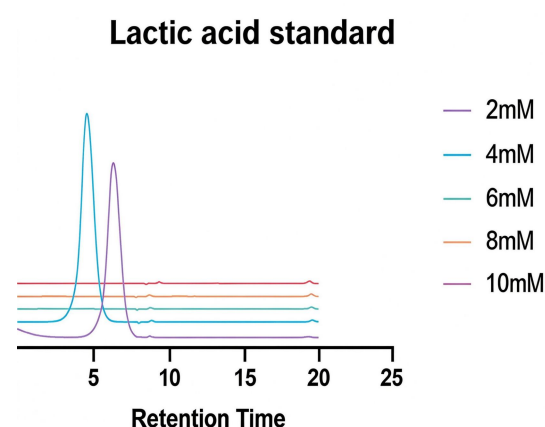


Figure 5 HPLC chromatograms and standard curve of lactic acid ($Y = 53.64X$, $R^2 = 0.9871$)

(D) HPLC results — full-pathway verification

Strain A-Y: pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE/BW. +NADH: 10 mM NADH (final). Empty vector: pYB1a-eGFP/BW. $\sqrt{}$: induced with arabinose; $\times$: not induced; the number denotes the whole-cell catalysis time (e.g., $\sqrt{1}/\sqrt{2}$: induced and catalysed for 1 h/2 h); 0: 0 h conversion fluid. The results are shown in Figure 6.

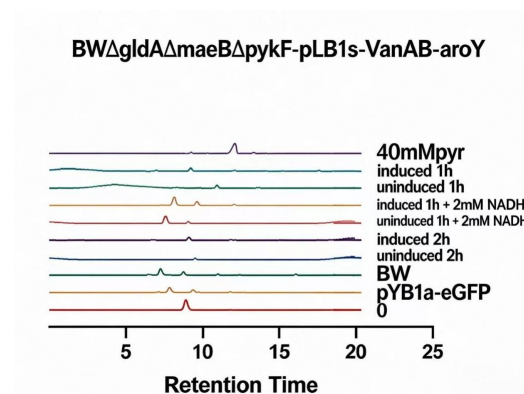


Figure 6 HPLC results of full-pathway verification

Results and analysis:

1. In the group without NADH, the substrate vanillic acid (VA) was consumed completely, whereas in the group with NADH, the full-pathway substrate VA was not consumed completely.
2. No obvious difference was observed between the induced and non-induced groups.
3. No peak appeared at the retention time of 15 min (no pyruvate production).

(E) HPLC results — full-pathway product verification

In the A-Y strain ($BW \Delta gldA \Delta maeB \Delta pykF$ -pLB1s-VanAB-aroY $BW \Delta gldA \Delta maeB \Delta pykF$ -pYB1a-dmpB--dmpD-mhpD-mhpE) that was induced and whole-cell catalysed for 1 h and 2 h in parallel systems, the samples were either not supplemented or supplemented with 10 mM pyruvate. No pyruvate was produced (Figure 7).

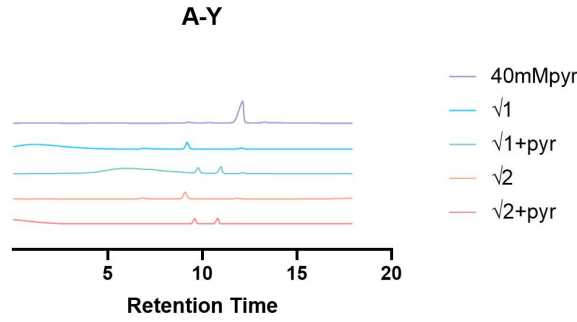


Figure 7 HPLC results of full-pathway product verification

(F) Pyruvate sensor characterization of the full pathway

1. 1×M9 medium preparation.

Antibiotic-free 1×M9 base medium (120 mL) was prepared as follows:

5×M9 salt solution	19.2 mL
30% glycerol	8.0 mL
1 M MgSO ₄	240 μL
Sterile ddH ₂ O	91.76 mL
1 M CaCl ₂	12 μL (added slowly last, mixed gently)

Vanillic-acid-containing 1×M9 (1 mM vanillic acid, 50 mL) was prepared as follows: 5.00 mL of 10 mM vanillic acid stock solution and 31.38 mL of sterile ddH₂O were mixed, and the pH was adjusted to 7.02 with 1 M NaOH. Then 9.60 mL of 5×M9 salt solution, 4.00 mL of 30% glycerol and 120 μL of 1 M MgSO₄ were added and mixed, and finally 6 μL of 1 M CaCl₂ was added gently (last).

2. Bacterial suspension treatment.

The strains that required no induction, namely the BW parent strain and the pyruvate sensor (bw-ΔgldA-ΔmaeB-ΔpykF), were shaken in LB medium for 16 h. The strain that required induction (pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE) was shaken and then induced in ZY medium for 16 h. For each strain, an OD = 6 bacterial suspension was taken into 1.5 mL EP tubes and centrifuged at 4200 rpm for 10 min; the supernatant was discarded, 600 μL of antibiotic-free 1×M9 was added and the cells were resuspended, followed by centrifugation at 4200 rpm for 10 min; the supernatant was discarded again, and 600 μL of antibiotic-free 1×M9 was added and resuspended.

3. Loading system.

The loading system is shown in Table 1. Seven groups were set up as follows:

Group	Medium	Inoculum (50 μL)	Additives
1: BW	Antibiotic-free 1×M9	BW suspension	—
2: Pyruvate sensor	Antibiotic-free 1×M9	Sensor suspension	—
3: Sensor + 2 g/L sodium pyruvate	Antibiotic-free 1×M9	Sensor suspension	2 g/L sodium pyruvate
4: Full pathway (induced) + VA (AS)	1×M9 containing VA	Induced full-pathway suspension	VA
5: Full pathway (not induced) + VA (AS)	1×M9 containing VA	Non-induced full-pathway suspension	VA
6: Full pathway (induced) blank (AS)	Antibiotic-free 1×M9	Induced full-pathway suspension	—
7: Full pathway (not induced) blank (AS)	1×M9 containing VA	Non-induced full-pathway suspension	—

Note: Group 1: 4950 μL of medium + 50 μL of suspension; Group 3: 3950 μL of medium + 1000 μL of 10 g/L sodium pyruvate stock + 50 μL of suspension.

4. Pyruvate sensor validation of the full pathway (induced group).

After induction of the pyruvate sensor co-transformed with pLB1s-VanAB-aroY and pYB1a-dmpB-dmpD-mhpD-mhpE, the growth curves of the seven groups were recorded (Figure 8). Under vanillic-acid-containing conditions, the induced and non-induced full-pathway groups were compared to exclude the influence of other metabolic pathways of vanillic acid.

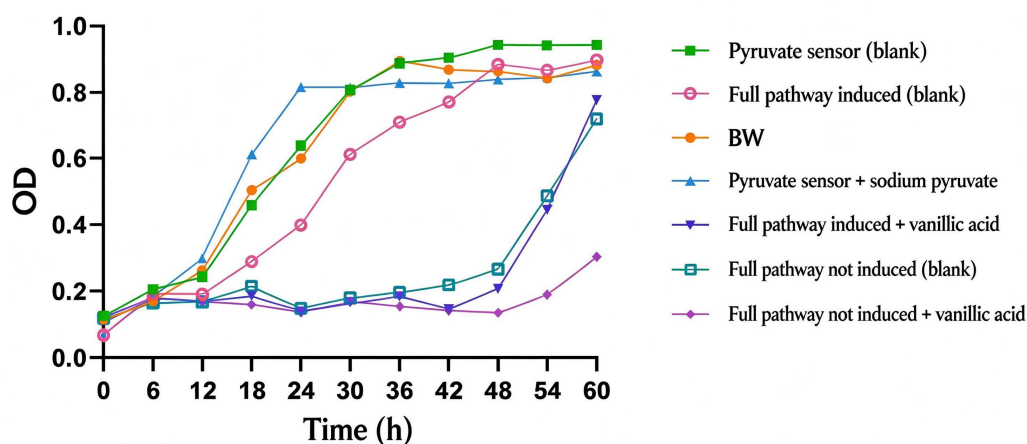


Figure 8 Growth curves of the pyruvate sensor for full-pathway validation

Week 12 (8.7-8.13)

Module 1:

(A) Verification of Expression Efficiency

1. Inoculation

Original strains: BW-pYB1a-eGFP、BW-pYB1a-、BW-pYB1a-nov1f、BW-pYB1a-progo

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-eGFP、BW-pYB1a-progo-nov1f、BW-pYB1a-nov1f、BW-pYB1a-progo

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

Using 4-PG to catalyze the induced enzymes for Isoeugenol production and Isoeugenol to catalyze the induced enzymes for vanillin production, vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Potassium phosphate buffer solution(PH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

Incubate in a shaker at 30°C for 1h ,3h, 6h, 9h hours.

4.Prepare mobile phase, standards and filter samples

Preparation of the mobile phase.

The mobile phase was prepared: Methanol: 0.1% acetic acid (ice) = 70:30.Methanol: 0.1% acetic acid (ice) = 60:40.Methanol: 0.1% acetic acid (ice) = 50:50.

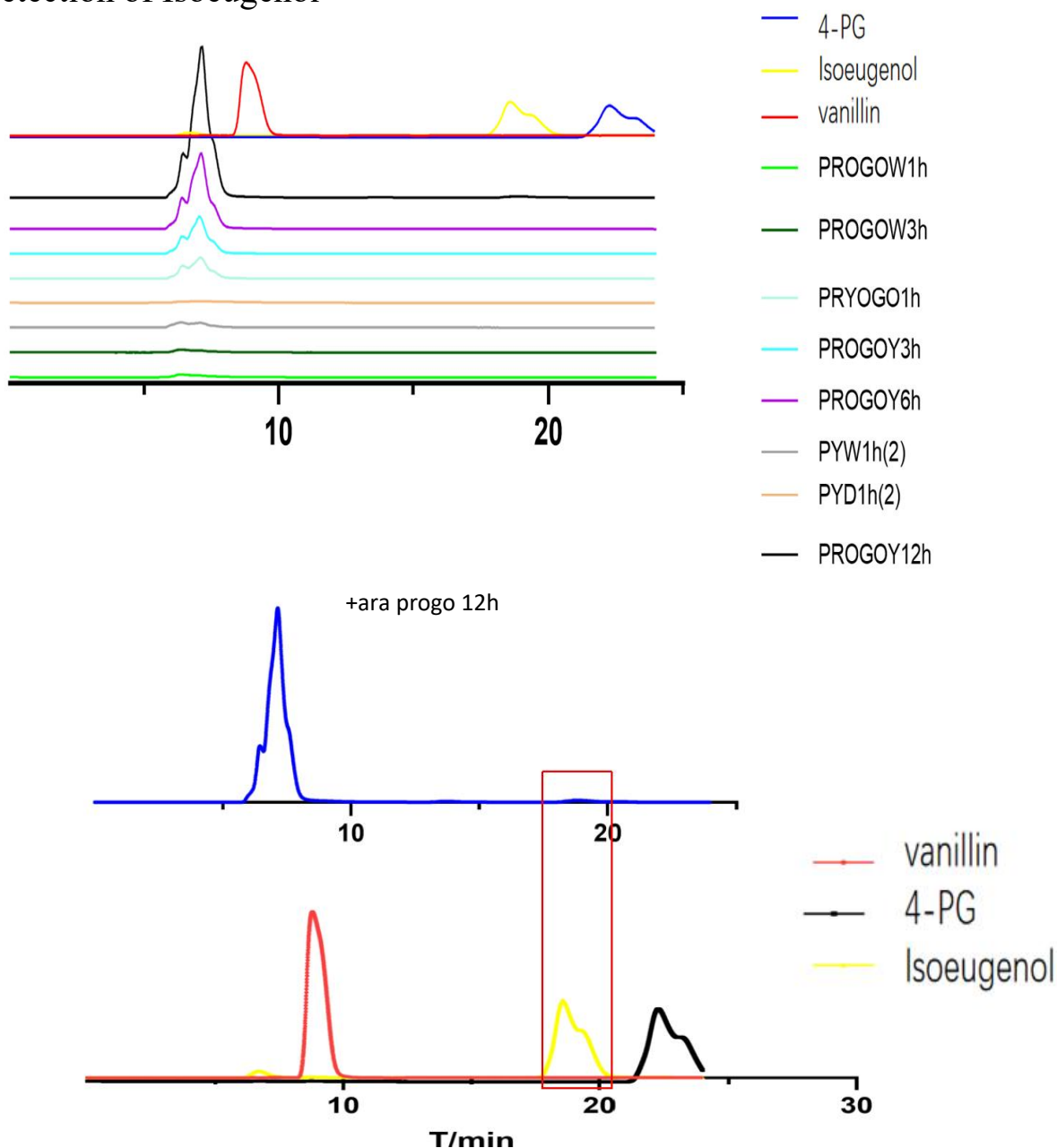
Preparation of standards and sample filtration.

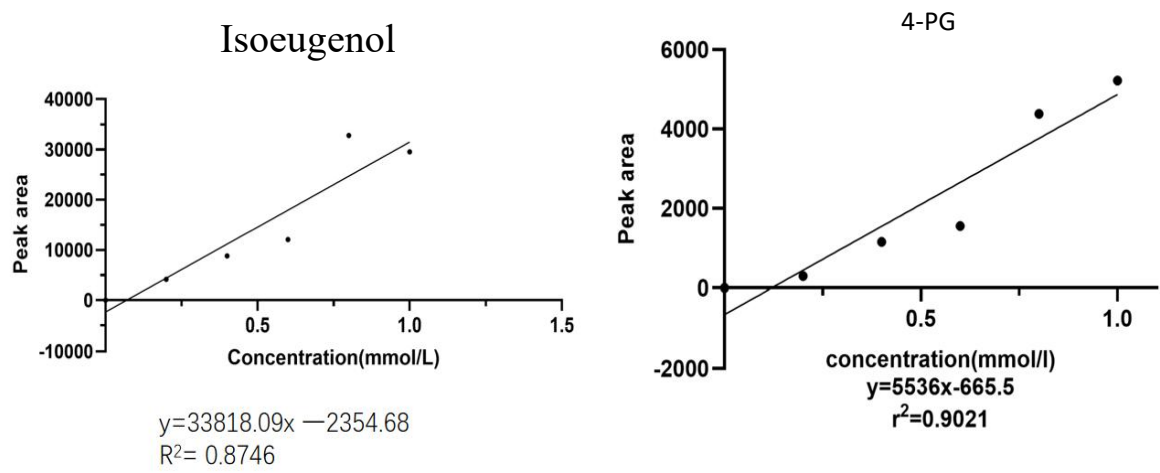
The standard solutions were prepared, and the samples were filtered through 0.22μm membranes for HPLC.

5.HPLC run (sulfuric acid method / phosphoric acid method)

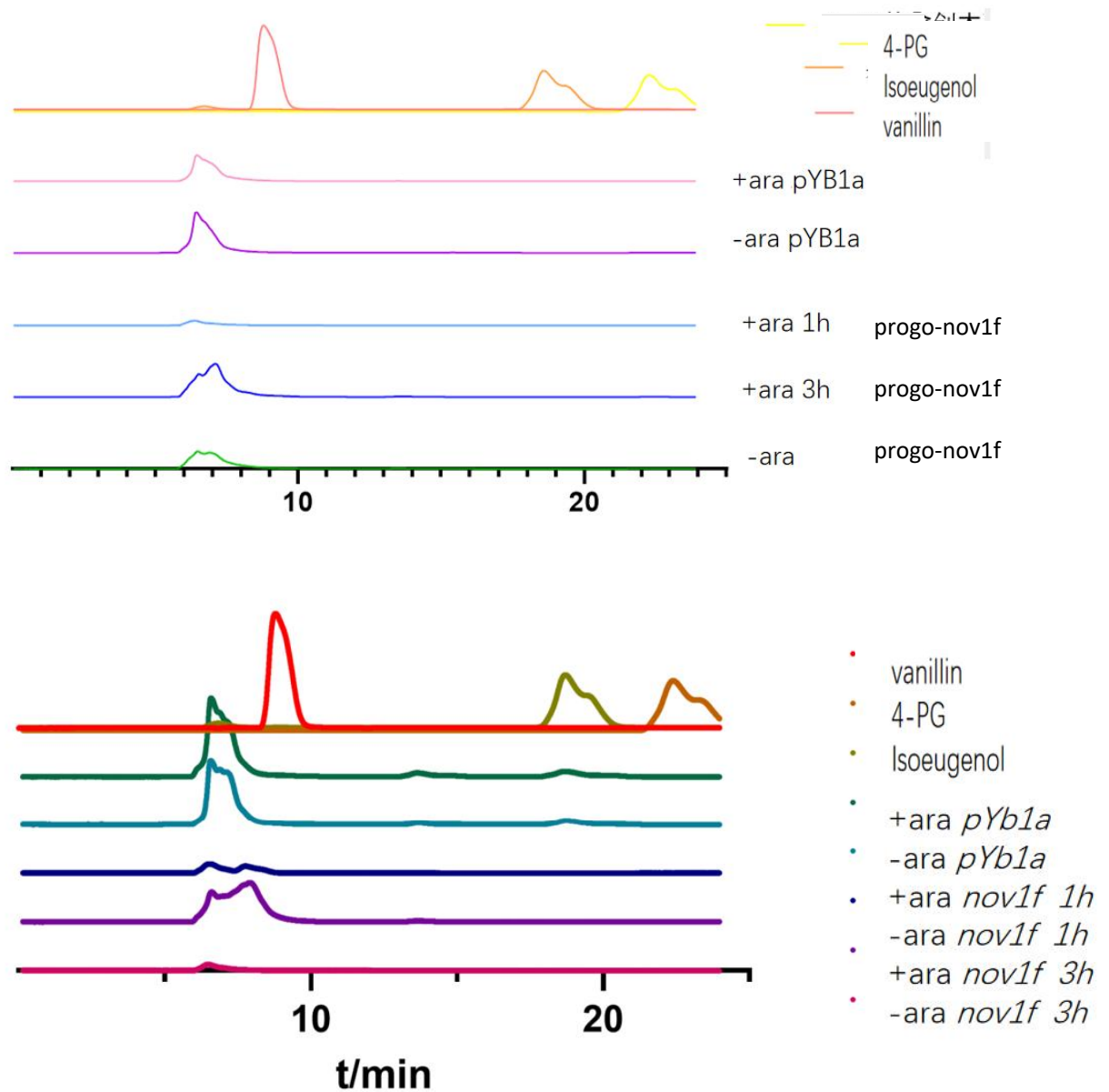
The samples were detected by HPLC using the sulfuric acid method (mobile phase: 5mM H₂SO₄) and the phosphoric acid method. Column: XP-C18; flow rate: 0.4 mL/min、 0.5 mL/min; injection volume: 10μL.

Detection of Isoeugenol





Detection of vanillin

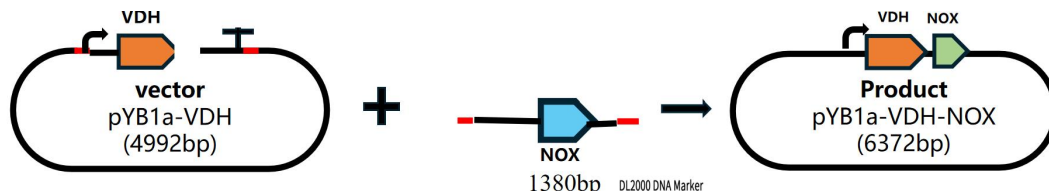


(B)Construct BW-pYB1a-*vdh-nox*

1.PCR + cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and enzymatic digestion of vector *pYB1a*

Subsequently, the relevant DNA fragments were assembled using Enzymatic cleavage ligation, followed by transformation into competent cells. Upon completion of the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed.

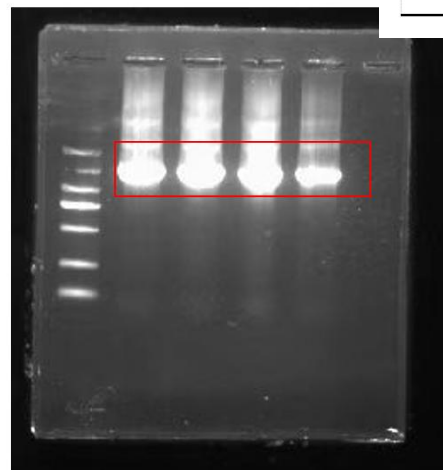
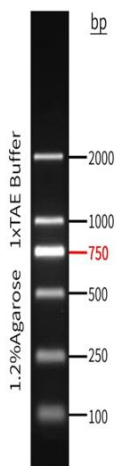


PCR system (50μL)	
2×Mix	25ul
<i>nox</i> -F	2ul
<i>nox</i> -R	2ul
Template	20ng
DDW	To 50ul

PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	41s
72°C	5min
25°C	∞

} × 30

DL2000 DNA Marker



3. Connection

Connecting the two fragments using the Enzymatic cleavage ligation.

3. Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



Module 2:

(A) Successful HPLC Detection of Wild-Type OMD Catalytic Activity Under Optimized Chromatographic Conditions

1. Induction

BW, pYB1a-eGFP, and pYB1a-ODM cultures grown at 37 °C for 12 h were taken out. For each group, a non-arabinose-induced control was set. Six tubes of ZY medium were brought into the laminar flow hood. Induction system was prepared as follows: 50×M 100 μL, 5052 100 μL, MgSO₄ 10 μL, 1000× stock 10 μL, antibiotic 5 μL, bacterial culture 50 μL, 20% arabinose 50 μL. Induced in a 30 °C shaker for 16 h.

2. Whole-cell catalysis

OD₆₀₀ was measured with a spectrophotometer and normalized to OD₆₀₀ = 6. Calculation: sampling volume (mL) = 6 ÷ measured OD₆₀₀. The bacterial suspension was aliquoted into 1.5 mL centrifuge tubes accordingly.

Tubes were carefully balanced and centrifuged at 4200 rpm for 10 min; all supernatant was discarded.

A 200 µL reaction system was prepared: vanillic acid (VA) at a final concentration of 2.5 mM in 50 mM Tris-HCl buffer (pH 7.5). The solution was added to each cell-pellet tube.

A 0 h control for each group was stored at -20 °C. Catalysis groups were incubated in a shaker at 30 °C, 200 r/min, protected from light, for 12 h.

3.Sample preparation

All post-catalysis samples were taken out and centrifuged at 13,000 rpm for 10 min to fully pellet the cells.

The clear supernatant was carefully pipetted.

The supernatant was filtered through a 0.22 µm aqueous filter membrane to remove impurities and collected in vials.

The filtrate was transferred into new disposable HPLC inserts, properly labeled, and placed into reusable HPLC sample vials.

All samples were stored at -20 °C pending HPLC injection.

4.Mobile phase preparation

Mobile phase was prepared at the ratio: 0.1% phosphoric acid aqueous solution : methanol = 75:25, mixed thoroughly. 10% methanol and pure methanol were also prepared, aliquoted into large bottles labeled with name and date.

This is an aqueous-organic mixed phase; the aqueous component was vacuum-filtered into the HPLC solvent reservoir to remove fine particles.

After filtration, the mobile phase was sonicated for 30 min for degassing to completely remove dissolved bubbles, preventing baseline instability and ghost peaks during HPLC runs.

5.HPLC Results

VA standard peak retention time: 21.2 min; 4-HB standard peak retention time: 17.8 min.

Actual sample peak retention time: 20.8 min; actual 4-HB peak retention time: 17.4 min.

HPLC conditions:

(1) Mobile phase: 0.1% phosphoric acid aqueous solution : methanol = 75:25

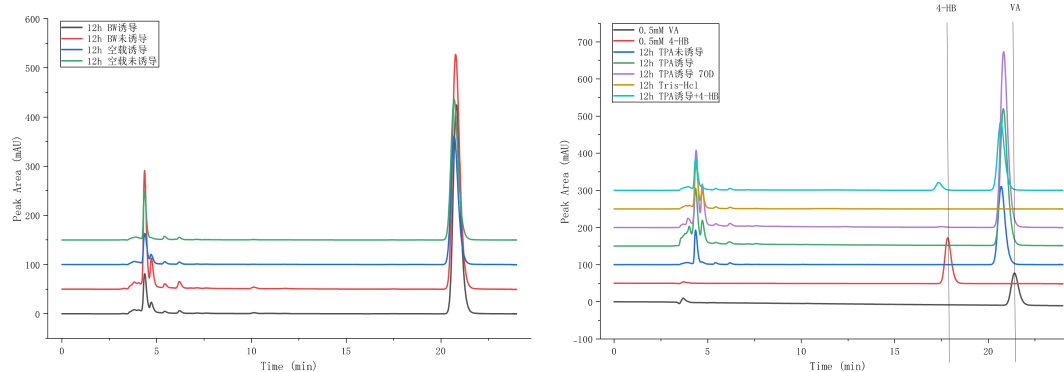
(2) Column: C18, 250 mm × 4.6 mm × 5 μm

(3) Detection wavelength: 254 nm

(4) Injection volume: 5 μL

(5) Flow rate: 0.7 mL/min

(6) Runtime per sample: 24 min



The catalytic activity of ODM was successfully detected.

(B) Construction of the *pYB1a-odm / gRB2k-pobR^{WT}-P_{pobA}*2-mCherry-CD-Cmr / BWΔCD*

Mutant Library

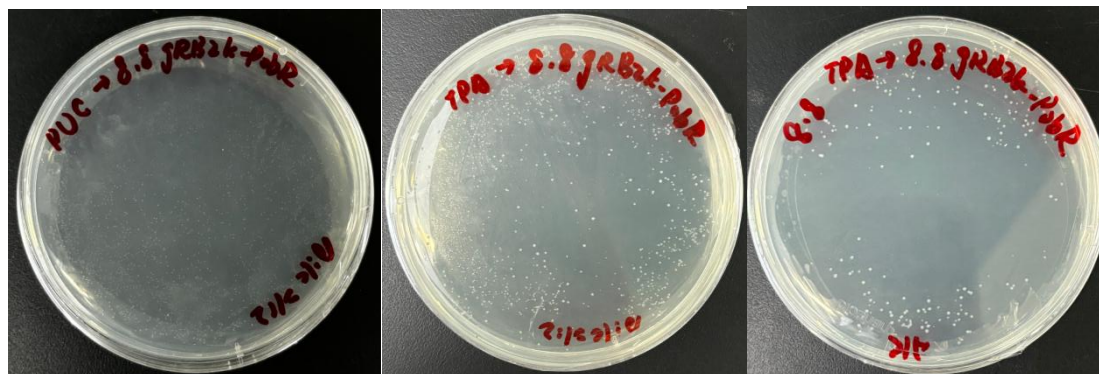
1.Re-extraction and Re-transformation of *gRB2k-pobR^{WT}-P_{pobA}*2-mCherry-CD-Cmr*

Plasmid; Re-preparation of Competent Cells

Incubated on ice for 30 min, heat-shocked at 42 °C for 45 s, then immediately placed on ice for 2–3 min.

After incubation, 600 μL LB was added to each EP tube and shaken at 37 °C for 50 min.

Centrifuged at 4200 rpm for 5 min, then spread on plates inside the biosafety cabinet.



- Transformation with 10 pg pUC19 control plasmid (efficiency check)

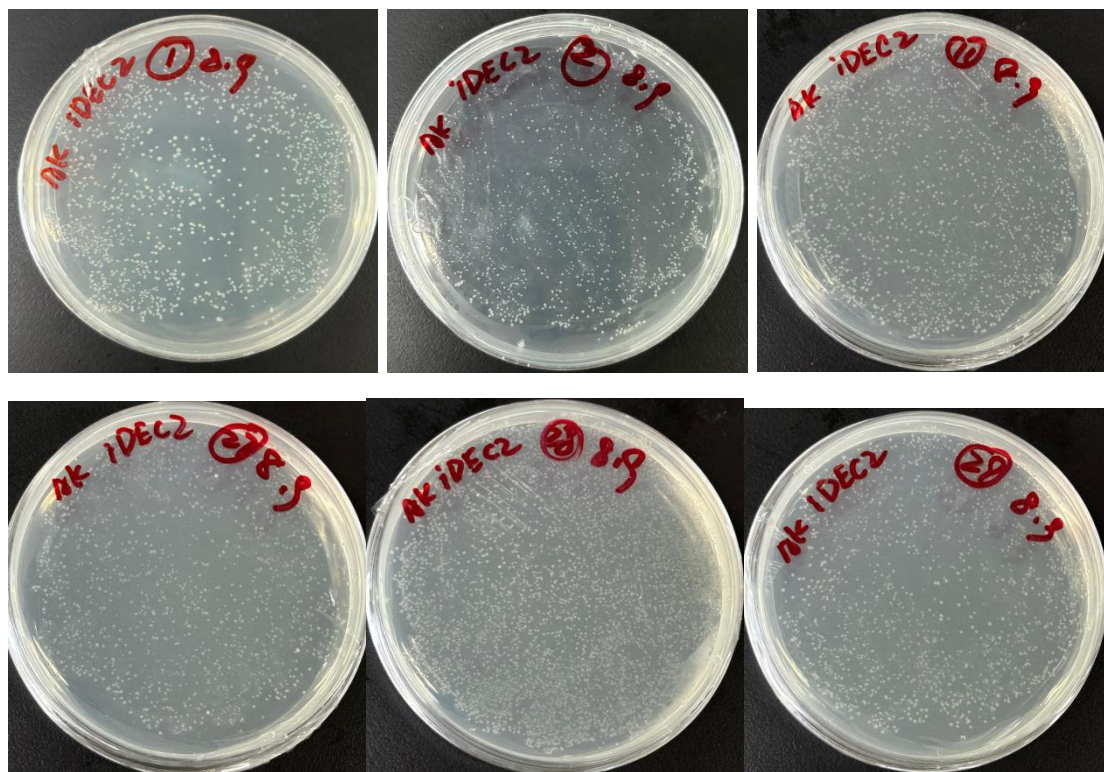
- Transformation with 20 ng ODM mixed plasmid; 50 μ L plated after centrifugation; A:K antibiotic ratio = 1:2

- Transformation with 20 ng ODM mixed plasmid; 50 μ L plated after centrifugation; A:K antibiotic ratio = 1:1

After 12 h of culture, the bacterial stratification phenomenon disappeared under normal antibiotic ratios, and colony morphology was normal; large-scale transformation for library construction is considered feasible.

2.Experiment Result 2 — Large-Scale Transformation: Construction of pYB1a-odm / gRB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr/ BW Δ CD Mutant Library

Mixed plasmid concentration: 20 ng/ μ L. Competent cells were incubated on ice for 5 min, then 1 μ L mixed plasmid was added. Incubated on ice for 30 min, heat-shocked at 42 $^{\circ}$ C for 45 s, then immediately placed on ice for 2–3 min. After incubation, 600 μ L LB was added to each EP tube and shaken at 37 $^{\circ}$ C for 50 min. Centrifuged at 4200 rpm for 5 min; 100 μ L was pipetted, mixed, and spread onto plates inside the biosafety cabinet.



Library capacity: 5×10^5 clones.

Colony scraping:

Collected plates cultured at 37 $^{\circ}$ C for 12 h. All operations performed inside the laminar flow hood. Added 5 mL antibiotic-containing LB liquid medium to each plate; discarded the remaining LB.

Used a sterile disposable inoculating loop to completely scrape all colonies from each plate, fully suspending the bacterial cells in the liquid.

Transferred all bacterial suspensions eluted from the plates into an empty conical flask using a pipette.

Sealed the conical flask and incubated with shaking at 37 $^{\circ}$ C for 6 h. Stored the bacterial culture after incubation.

(C) Biosensor-Based M9 Selection Assay on the pYB1a-ODM Mutant Library

1. Screening Assay on the pYB1a-ODM / gRB2k-*pobR*^{WT}-P_{pobA*2}-*mCherry-CD-Cmr*/ BW Δ CD Mutant Library

M9 selection system was set up. Bacterial culture, vanillic acid, and chloramphenicol concentrations were tested:

(1) VA concentration: 2.5 mM (10 mM stock; diluted pH = 3.3, adjusted to ~pH 7 with NaOH)

(2) Bacterial culture volume: 50 μ L (1% inoculum)

(3) Chloramphenicol concentration: 37.5 mg/L

Detailed system (5 mL)				
Component	WT group	pobR group	BW Δ CD group	Mixed (AK) group
Antibiotic	A 5 μ L	K 5 μ L	None	A 5 μ L, K 5 μ L
5 $\times$ M9	1 mL	1 mL	1 mL	1 mL
20% Glycerol	500 μ L	500 μ L	500 μ L	500 μ L
MgSO ₄	10 μ L	10 μ L	10 μ L	10 μ L
CaCl ₂	0.5 μ L	0.5 μ L	0.5 μ L	0.5 μ L
20% Arabinose	50 μ L	50 μ L	50 μ L	50 μ L
Chloramphenicol (C)	5 μ L	5 μ L	5 μ L	5 μ L
Water + VA system	3372 μ L	3372 μ L	3377 μ L	3367 μ L
Bacteria	50 μ L	50 μ L	50 μ L	50 μ L

Module 3:

(A) HPLC sample preparation - induction and whole-cell catalysis (0 h, 2 h)

1. Induction

Strains: (1) BW; (2) BW/pLB1s-VanAB-aroY, pYB1a-dmpBD-mhpDE; (3) pYB1a-dmpBD-mhpDE; (4) BW/pYB1a-eGFP.

Induction system (5 mL):

Component	Volume
50x M	100 μ L
5052	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L

Antibiotic	5 uL
Bacterial suspension	50 uL
20% arabinose induction solution	50 uL

Incubated at 30 C with shaking for 16 h.

2. Whole-cell catalysis

Sample volume (mL) = 6 / measured OD600.

Incubated at 30 C with shaking for 2 h.

(B) HPLC preparation - mobile phase configuration and method

1. Mobile phase configuration

5 mM sulfuric acid aqueous solution, filtered and ultrasonicated for 30 min.

5% acetonitrile + 5 mM sulfuric acid; 30% acetonitrile + 5 mM sulfuric acid, filtered and ultrasonicated for 30 min.

2. HPLC method

Parameter	Value
Mobile phase	5 mM sulfuric acid aqueous solution
Flow rate	0.6 mL/min
Column temperature	50 C
Column type	Aminex HPX-87H
Injection volume	10 uL

3. Result

The needle seat was clogged, causing the pressure to exceed the column's maximum tolerance. Multiple attempts failed, so the run was stopped. No HPLC results were obtained.

(C) Pyruvate sensor characterization and validation

1. Colony PCR verification

Checking BW delta gldA delta maeB delta pykF for contamination. Target band: 1635 bp; colony PCR verified correct; single colonies were picked and expanded for shaking culture.

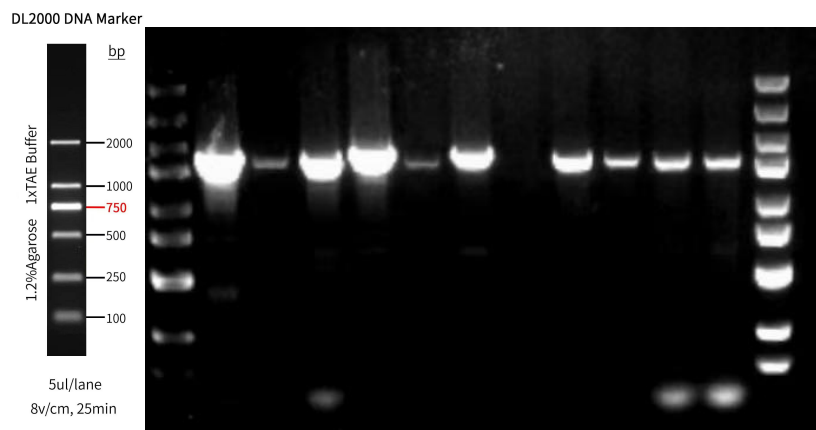


Figure 1. Colony PCR verification of BW delta gldA delta maeB delta pykF (target band 1635 bp).

2. 1x M9 medium preparation

Glycerol final concentration 0.05%, reduced carbon source.

1x M9 antibiotic-free base solution (90 mL): 18 mL 5x M9 salt solution, 150 uL 30% glycerol, 180 uL 1 M MgSO₄, 71.66 mL sterile ddH₂O; finally slowly add 9 uL 1 M CaCl₂ and mix gently.
 1x M9 with 1 mM vanillic acid (50 mL): 6.50 mL 10 mM vanillic acid stock, 30.00 mL sterile ddH₂O, adjust pH to 7.2 with 1 M NaOH; then add 13.00 mL 5x M9 salt solution, 108 uL 30% glycerol, 130 uL 1 M MgSO₄, mix; AS antibiotic 65 uL, 6.5 uL 1 M CaCl₂, mix gently; add 15.19 mL sterile ddH₂O.

3. Bacterial suspension treatment

Reduced inoculum. BW, BW delta gldA delta maeB delta pykF, and BW delta gldA delta maeB delta pykF / pLB1s-VanA-VanB-aroY, pYB1a-dmpBD-mhpDE were cultured in LB medium. Take OD₆₀₀ = 6.0, centrifuge and discard supernatant.

Master tube A (for 96-well plate 1% inoculum): add 600 uL antibiotic-free 1x M9, resuspend by pipetting, centrifuge, discard supernatant; repeat 3 times to obtain master tube A (OD₆₀₀ = 6).

Working tube B (for tube group 0.01% inoculum): 150 uL master tube A + 450 uL ordinary antibiotic-free 1x M9, mix (OD₆₀₀ = 1.5).

Working tube C (for 96-well plate 0.01% inoculum): 6 uL master tube A + 594 uL ordinary antibiotic-free 1x M9, mix (OD₆₀₀ = 0.06).

4. Loading system - tube group

5 mL system, 0.01% inoculum.

Group	1x M9 medium (uL)	Additional reagent (uL)	Working tube B (uL)
1 BW	4998 ordinary AF 1x M9	-	2
2 Pyruvate sensor	4998 ordinary AF 1x M9	-	2
3 Sensor + 2 g/L sodium pyruvate (18.2 mM)	3998 ordinary AF 1x M9	10 g/L sodium pyruvate stock 1000	2
4 Full pathway induced + vanillic acid (AS)	4948 1x M9 + vanillic acid	Arabinose stock 50	2
5 Full pathway not induced + vanillic acid (AS)	4998 1x M9 + vanillic acid	-	2
6 Full pathway induced blank	4948 ordinary AF 1x M9	Arabinose stock 50	2
7 Full pathway not induced blank	4998 ordinary AF 1x M9	-	2

5. Loading system - 96-well plate group

0.01% inoculum: 2 uL working tube C; 1% inoculum: 2 uL master tube A.

Group	1x M9 medium (uL)	Additional reagent (uL)	Bacterial suspension (uL)
1 BW	198 ordinary AF 1x	-	2

	M9		
2 Pyruvate sensor	198 ordinary AF 1x M9	-	2
3 Sensor + 2 g/L sodium pyruvate	158 ordinary AF 1x M9	10 g/L sodium pyruvate stock 40	2
4 Full pathway induced + vanillic acid (AS)	196 1x M9 + vanillic acid	Arabinose stock 2	2
5 Full pathway not induced + vanillic acid (AS)	198 1x M9 + vanillic acid	-	2
6 Full pathway induced blank	196 ordinary AF 1x M9	Arabinose stock 2	2
7 Full pathway not induced blank	198 ordinary AF 1x M9	-	2

6. Results and analysis

After reducing the carbon source concentration, neither group grew (both tube and 96-well plate groups).

After 48 h, leakage occurred in the 96-well plate, and measurement was terminated.

After reducing both carbon source and inoculum, the bacteria did not grow, and the differentiation was minimal.

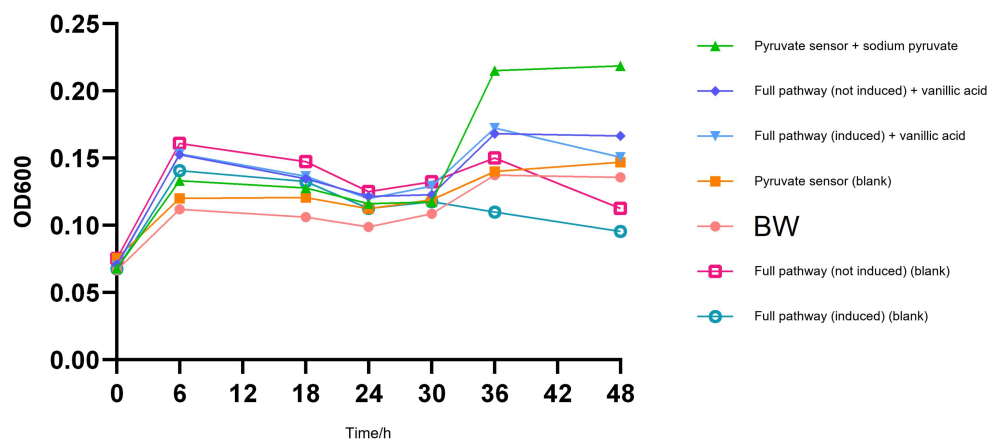


Figure 2. Growth curve of the 96-well plate group at 1% inoculum.

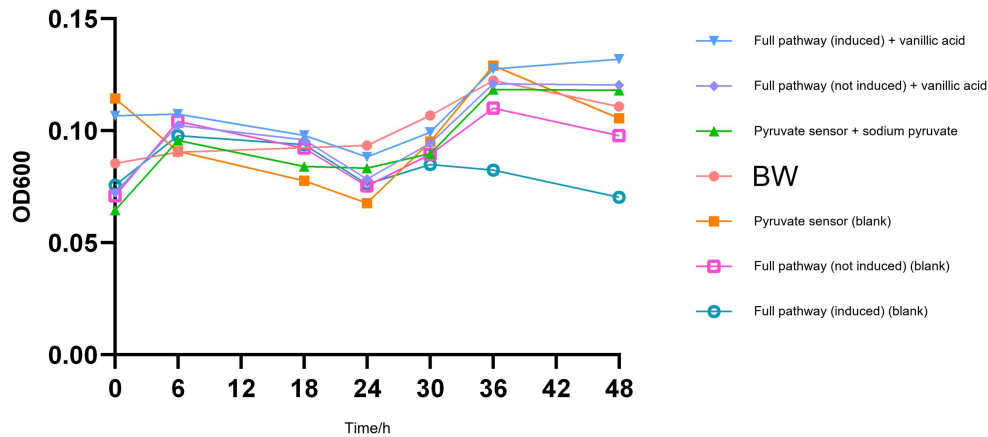


Figure 3. Growth curve of the 96-well plate group at 0.01% inoculum.

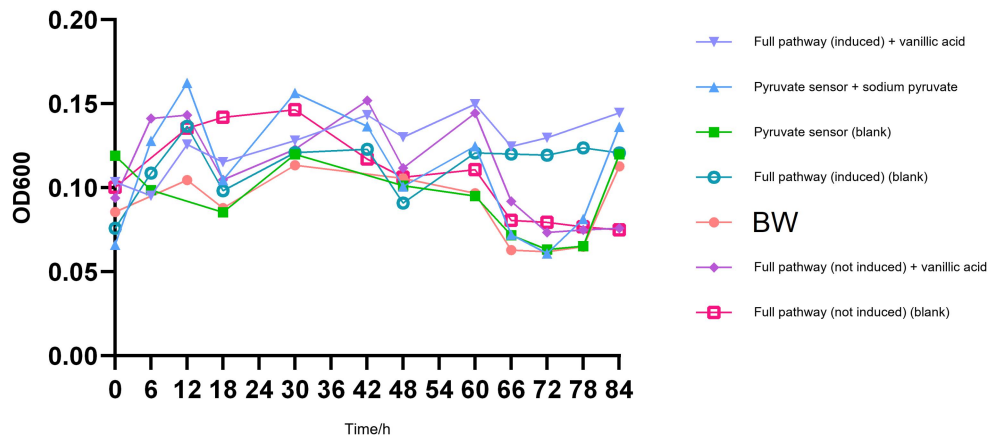


Figure 4. Growth curve of the tube group at 0.01% inoculum.

Week 13 (8.14-8.19)

Module 1:

1. Inoculation

Original strains : BW-pYB1a-eGFP 、 BW-pYB1a-progo-nov1f 、 BW-pYB1a-nov1f 、
BW-pYB1a-progo、 BW-pYB1a-vdh

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-eGFP、 BW-pYB1a-progo-nov1f、 BW-pYB1a-nov1f、 BW-pYB1a-progo

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

Using 4-PG to catalyze the induced enzymes for Isoeugenol production and Isoeugenol to catalyze the induced enzymes for vanillin production, vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Tris-HCl(pH=7.5) or 50mM Potassium phosphate buffer solution(PH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

Incubate in a shaker at 30°C for 1h, 3h hours.

4. Prepare mobile phase, standards and filter samples

Preparation of the mobile phase.

The mobile phase was prepared: Methanol: 0.1% acetic acid (ice) = 70:30. Methanol: 0.1% acetic acid (ice) = 65:35.

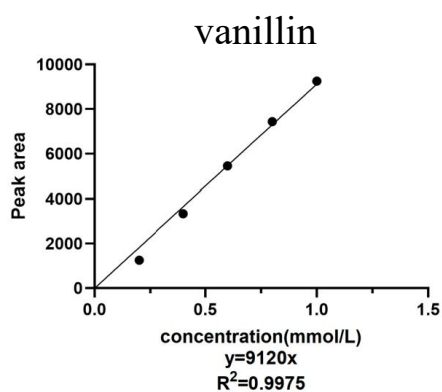
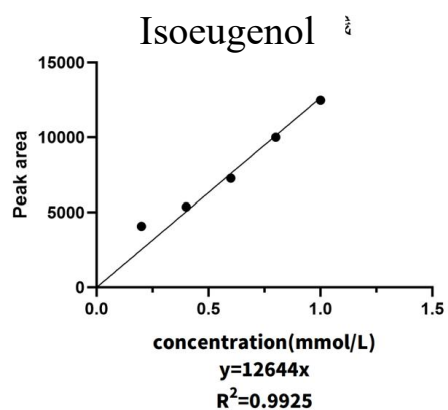
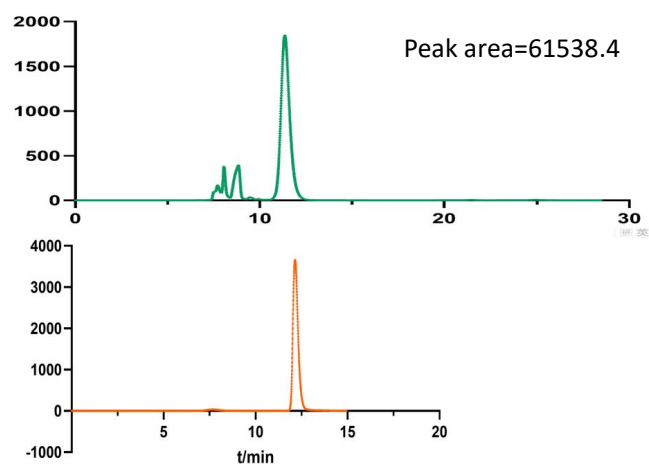
Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

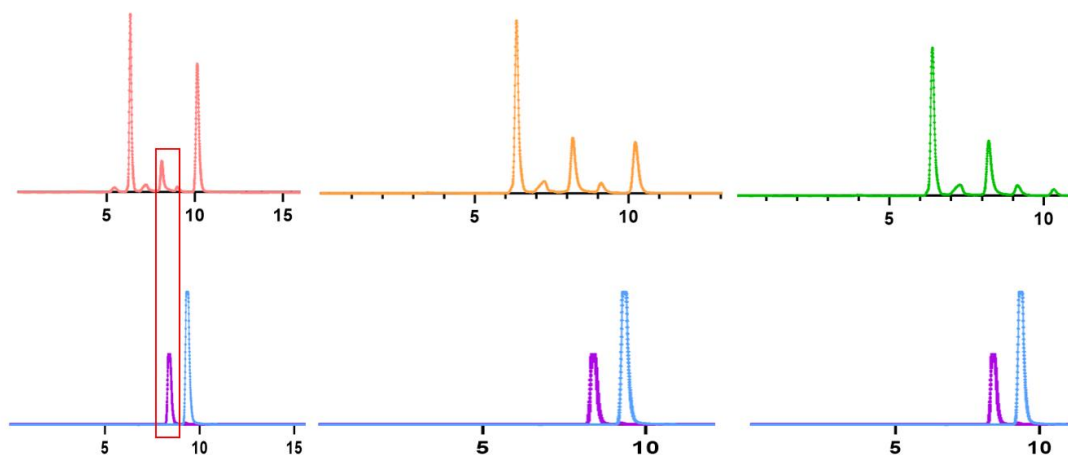
5. HPLC run (sulfuric acid method / phosphoric acid method)

The samples were detected by HPLC using the sulfuric acid method (mobile phase: 5mM H₂SO₄) and the phosphoric acid method. Column: XP-C18; flow rate: 0.4 mL/min 0.5 mL/min; injection volume: 10μL.

Detection of vanillin

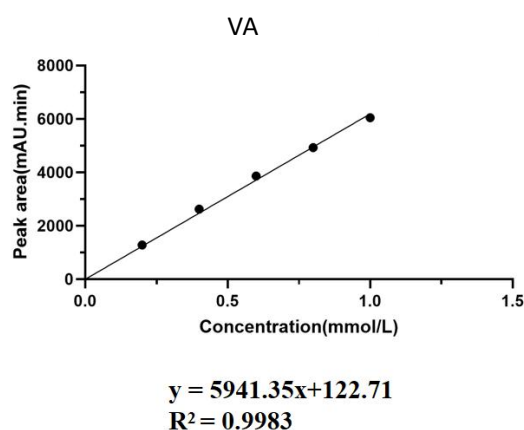
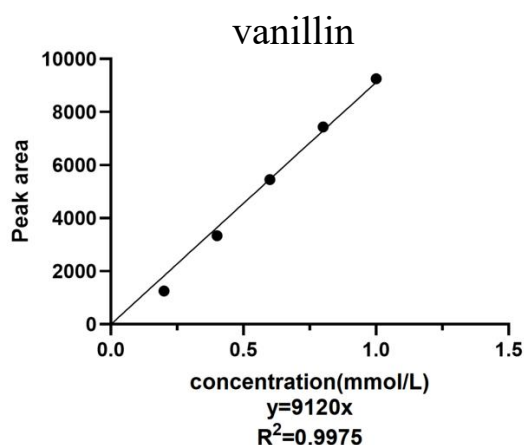


Detection of VA



After dilution by ten times, the one-hour sealing area was 1124.43, the six-hour sealing area was 1811.4, and the six-hour conversion rate was 2338.2.

The one-hour conversion rate was 41.13%, the six-hour conversion rate was 66.26%, and the 15-minute conversion rate was 85%.

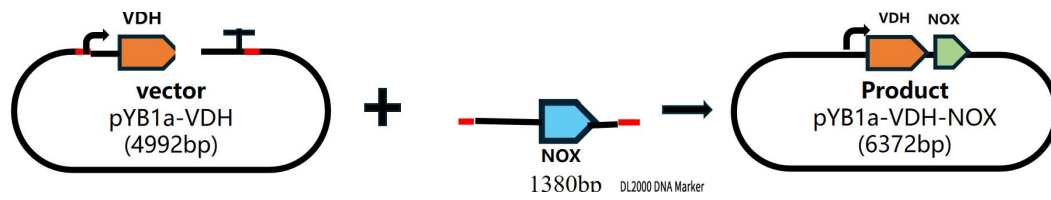


(B)Construct BW-pYB1a-*vdh-nox*

1.PCR + cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and enzymatic digestion of vector *pYB1a*

Subsequently, the relevant DNA fragments were assembled using Enzymatic cleavage ligation, followed by transformation into competent cells. Upon completion of the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed.

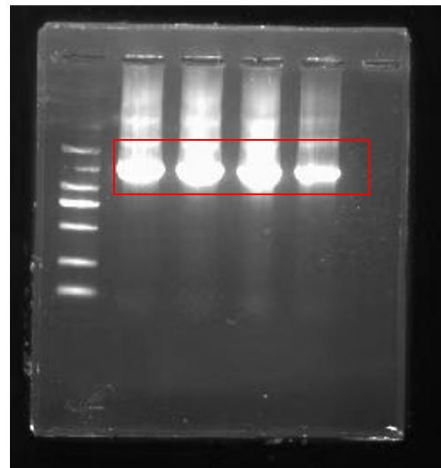
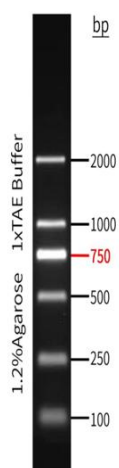


PCR system (50μL)	
2×Mix	25ul
<i>nox</i> -F	2ul
<i>nox</i> -R	2ul
Template	20ng
DDW	To 50ul

PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	41s
72°C	5min
25°C	∞

} × 30

DL2000 DNA Marker

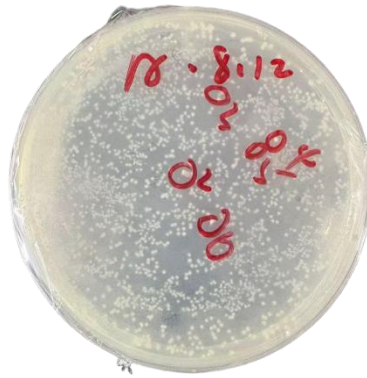


4. Connection

Connecting the two fragments using the Enzymatic cleavage ligation.

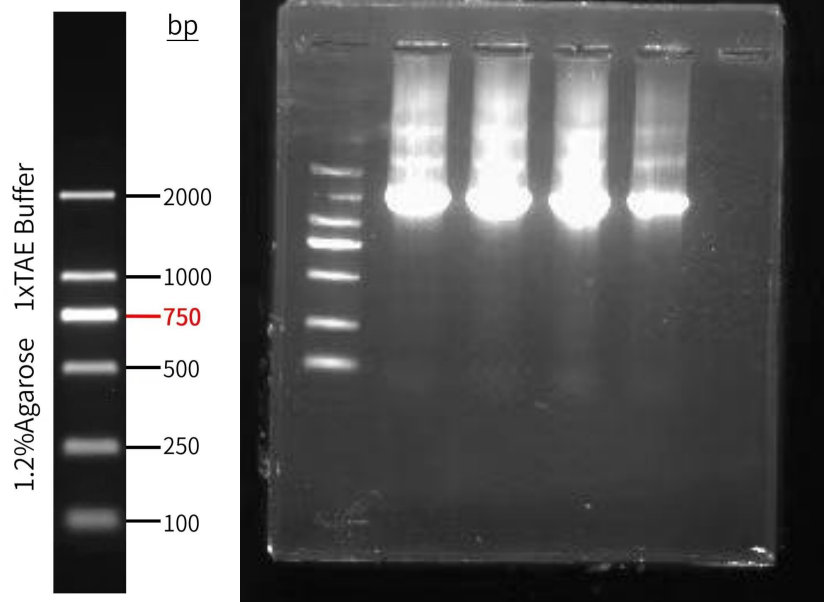
3. Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



4.Colony PCR

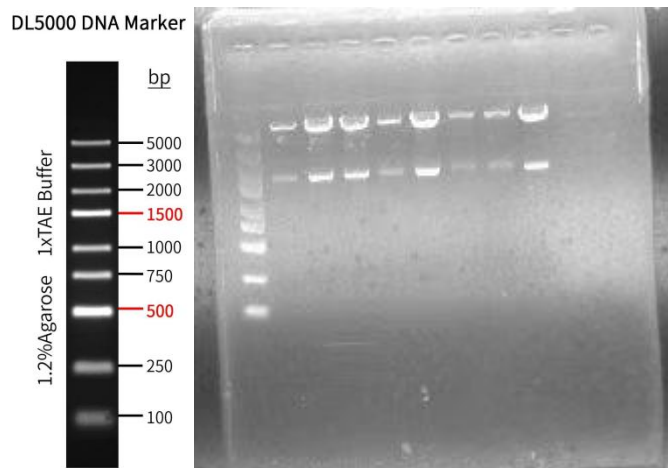
NOXF/PBR was chosen as the primer with a theoretical length of 1384bp
DL2000 DNA Marker



5.Enzyme digestion

Selected 10 colonies for using *SpeI* *PstI* to verify the BW-pYB1a-*vdh-nox*. Enzyme digestion system:

Enzyme digestion (10μL)	
<i>SpeI</i>	0.2μL
<i>PstI</i>	0.2μL
rCutsmart	1.0μL
质粒	200ng
DDW	to 10μL



The products were analyzed by agarose gel electrophoresis. The results showed that the enzyme digestion was right.

6.Plasmid extraction + send for sequencing

The correctly digested protoplasmid was sent for testing, and BW-pYB1a-*vdh-nox* was sequenced correctly.

Module 2:

(A) Optimization of M9 Biosensor Screening System and Contamination Diagnosis of the Mutant Library

1.Screening Assay on the pYB1a-ODM / gRB2k-*pobR*^{WT}-P_{pobA*2}-*mCherry-CD-Cmr*/ BW Δ CD Mutant Library

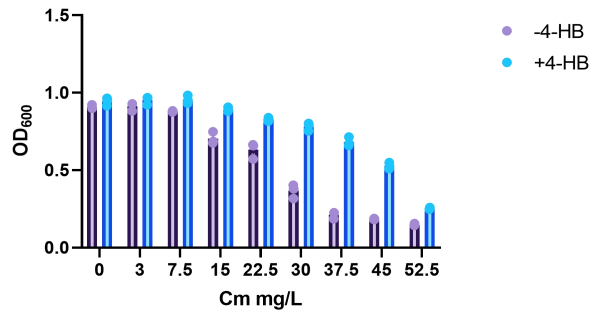
M9 selection system was set up. Bacterial culture, vanillic acid, and chloramphenicol concentrations were tested:

- (1) VA concentration: 2.5 mM (10 mM stock added directly, diluted with water to a final concentration of 2.5 mM)
- (2) Bacterial culture volume: 50 μ L (1% inoculum)
- (3) Chloramphenicol concentration: 37.5 mg/L

Detailed system (5 mL)				
Component	WT(<i>pobR</i>) group	<i>pobR</i> group	BW Δ CD group	Mixed (AK) group
Antibiotic	A 5 μ L, K 5 μ L	K 5 μ L	None	A 5 μ L, K 5 μ L

Detailed system (5 mL)				
Component	WT(pobR) group	pobR group	BWΔCD group	Mixed (AK) group
5× M9	1 mL	1 mL	1 mL	1 mL
20% Glycerol	500 μL	500 μL	500 μL	500 μL
MgSO ₄	10 μL	10 μL	10 μL	10 μL
CaCl ₂	0.5 μL	0.5 μL	0.5 μL	0.5 μL
20% Arabinose	50 μL	50 μL	50 μL	50 μL
Chloramphenicol (C)	12.5 μL	12.5 μL	12.5 μL	12.5 μL
10 mM VA	1250 μL	1250 μL	1250 μL	1250 μL
Water	2117 μL	2122 μL	2127 μL	2117 μL
Bacteria	50 μL	50 μL	50 μL	50 μL

梯度Cm浓度下携带gYB2a-PobR^{WT}-P_{pobA}*2-mCherry-codA-cmr菌株生长表现 (t=10h)



M9 selection system was set up. Bacterial culture, vanillic acid, and chloramphenicol concentrations were tested:

- (1) VA concentration: 1 mM (10 mM stock added directly, diluted with water to a final concentration of 1 mM)
- (2) Bacterial culture volume: 50 μL (1% inoculum)
- (3) Chloramphenicol concentration: 45 mg/L

Detailed system (5 mL)

Component	WT(pobR) group	pobR group	BWΔCD group	Mixed (AK) group
Antibiotic	A 5 μL, K 5 μL	K 5 μL	None	A 5 μL, K 5 μL
5× M9	1 mL	1 mL	1 mL	1 mL
20% Glycerol	500 μL	500 μL	500 μL	500 μL
MgSO ₄	10 μL	10 μL	10 μL	10 μL
CaCl ₂	0.5 μL	0.5 μL	0.5 μL	0.5 μL
20% Arabinose	50 μL	50 μL	50 μL	50 μL
Chloramphenicol (C)	15 μL	15 μL	15 μL	15 μL
VA system	500 μL	500 μL	500 μL	500 μL
Water	2864.5 μL	2869.5 μL	2874.5 μL	2864.5 μL
Bacteria	50 μL	50 μL	50 μL	50 μL

After confirming the screening workflow and system, results after 12 h of culture:



Groups from left: BWΔCD, pobR, WT(pobR), library + VA, library without VA.

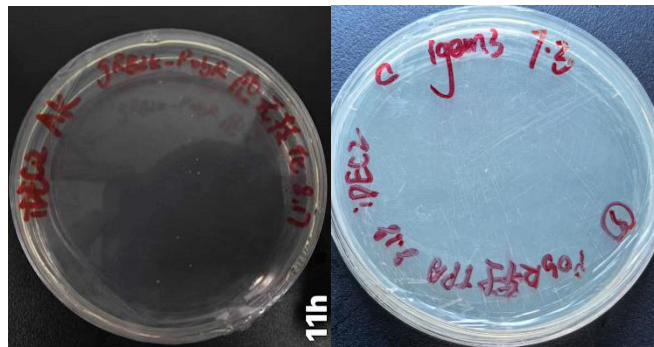
The no-substrate control groups still grew, suggesting contamination. The experiment was repeated to verify whether the mutant library was contaminated.



From left: mutant library + ACK, WT(pobR), library + VA, library without VA; then WT(pobR), library + VA, library without VA.

- Left panel: VA final concentration 2.5 mM; chloramphenicol 37.5 mg/L

- Right panel: VA final concentration 1 mM; chloramphenicol 45 mg/L

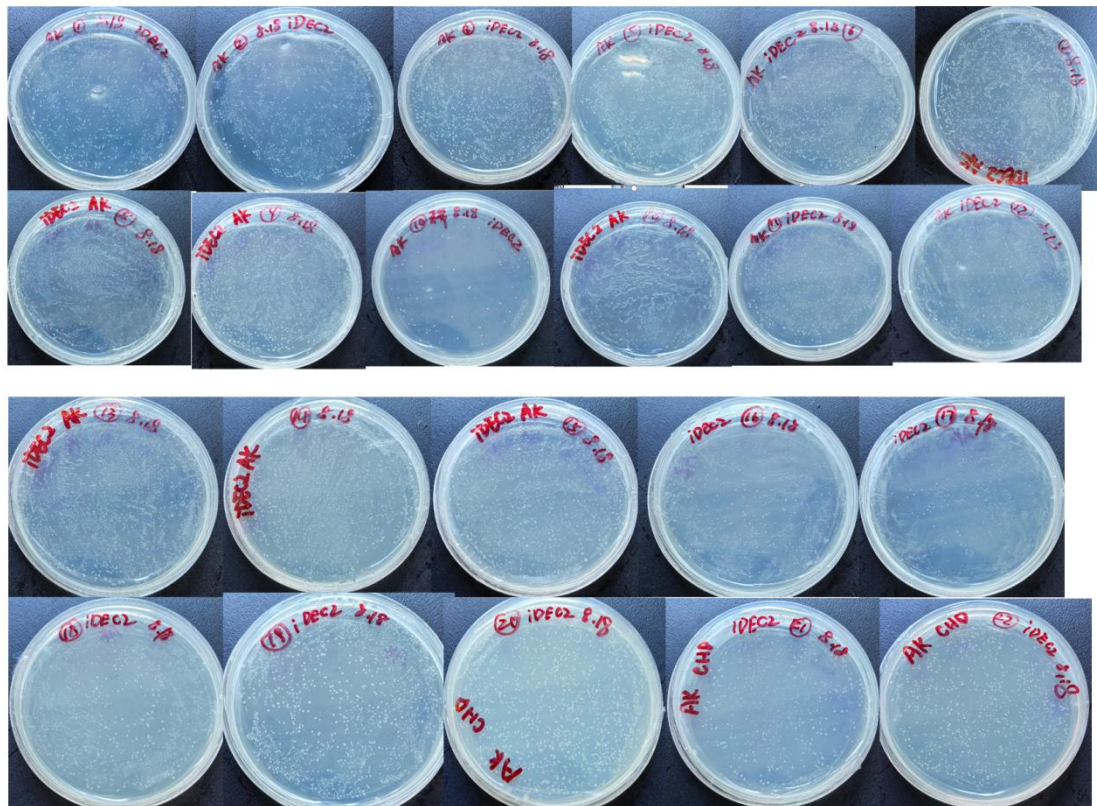


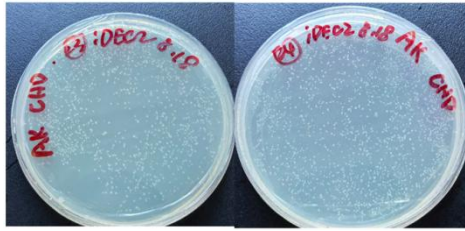
pobR empty competent cells on AK plate: no growth.

Mixed library on C-antibiotic plate: no growth.

(B) Reconstruction of the pYB1a-ODM / gRB2k-pobR^{WT}-P_{pobA}*2-mCherry-CD-CmrI BWΔCD Mutant Library After Contamination Removal

Mixed plasmid concentration: 20 ng/μL. Competent cells were incubated on ice for 5 min, then 1 μL mixed plasmid was added. Incubated on ice for 30 min, heat-shocked at 42 °C for 45 s, then immediately placed on ice for 2–3 min. After incubation, 600 μL LB was added to each EP tube and shaken at 37 °C for 50 min. Centrifuged at 4200 rpm for 5 min; 100 μL was pipetted, mixed, and spread onto plates inside the biosafety cabinet.





Library capacity: 4×10^4 clones.

Colony scraping:

Collected plates cultured at 37 °C for 12 h. All operations performed inside the laminar flow hood. Added 5 mL antibiotic-containing LB liquid medium to each plate; discarded the remaining LB.

Used a sterile disposable inoculating loop to completely scrape all colonies from each plate, fully suspending the bacterial cells in the liquid.

Transferred all bacterial suspensions eluted from the plates into an empty conical flask using a pipette.

Sealed the conical flask and incubated with shaking at 37 °C for 5 h. Stored the bacterial culture after incubation.

(C) Contamination Re-Verification and Restarted Biosensor Screening of the Rebuilt Library

1. Screening Assay on the pYB1a-ODM / gRB2k-*pobR*^{WT}-P_{pobA*2}-*mCherry-CD-Cmr*/ BW Δ CD Mutant Library

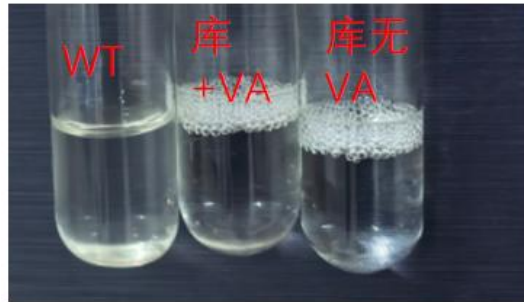
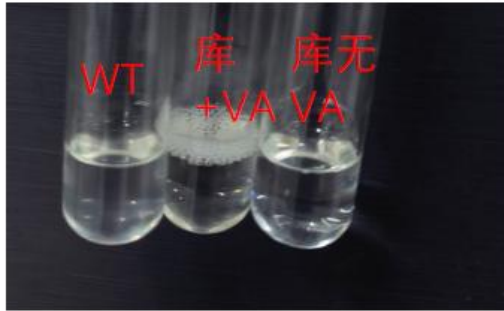
The mutant library was inoculated into LB with A, C, and K antibiotics separately for contamination check.

Results after 11 h of culture:



- A antibiotic - K antibiotic - C antibiotic

Screening results after 11 h of culture:



- System 1: VA 2.5 mM, Cmr 37.5 mg/L

- Library + VA

- Library without VA

- System 2: VA 1 mM, Cmr 45 mg/L

- Library + VA

- Library without VA

- WT controls

The growth pattern is considered normal with no contamination observed; cultures were returned to the 30 °C shaker for continued incubation

Module 3:

(A) Pyruvate sensor colony PCR verification

Target bands: 1380 bp, 1159 bp, 1635 bp.

Accuracy rate = $10/11 \times 4/5 \times 3/5$ approx 43.6%.

DL2000 DNA Marker

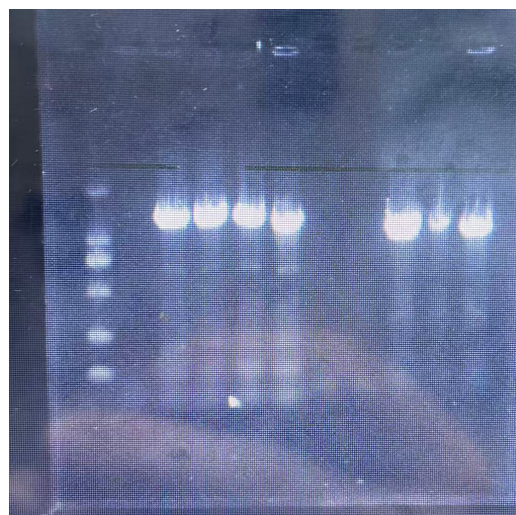
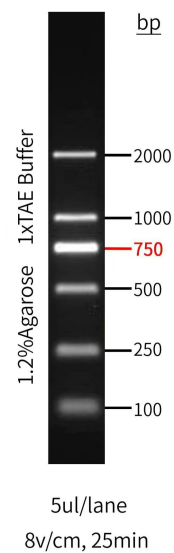


Figure 5. Colony PCR verification of the pyruvate sensor (target bands 1380 bp, 1159 bp, 1635 bp).

(B) Plasmid construction - gR9s-bio177-mCherry-CD-CmR

1. Replication origin adjustment (gY9s to gR9s)

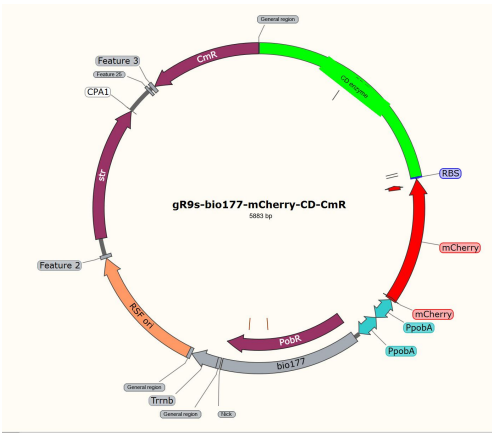


Figure 6. Circular plasmid map of gR9s-bio177-mCherry-CD-CmR after replication origin adjustment.

2. Colony PCR result - correct

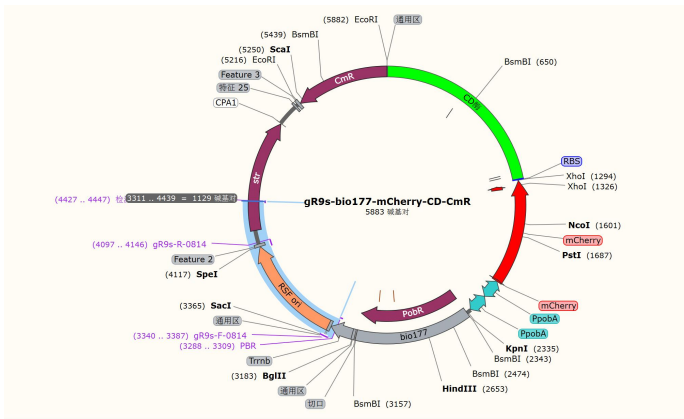
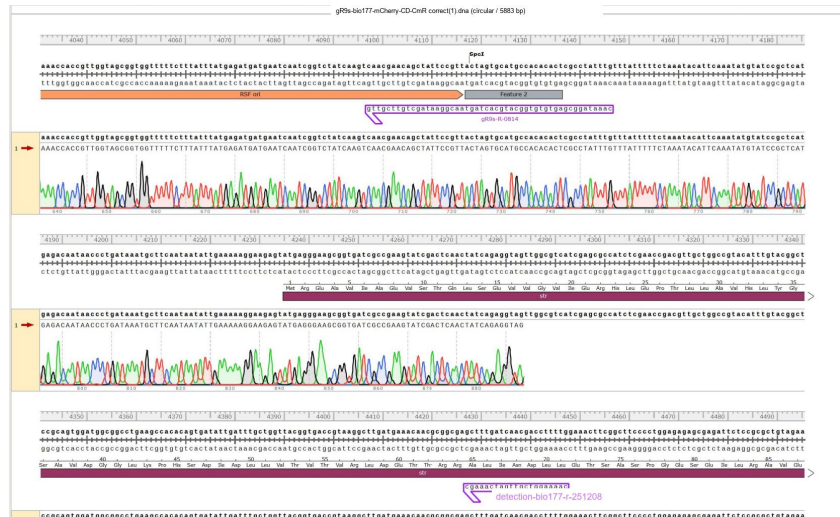


Figure 7. Colony PCR result of gR9s-bio177-mCherry-CD-CmR (correct).

3. Sequencing result - correct

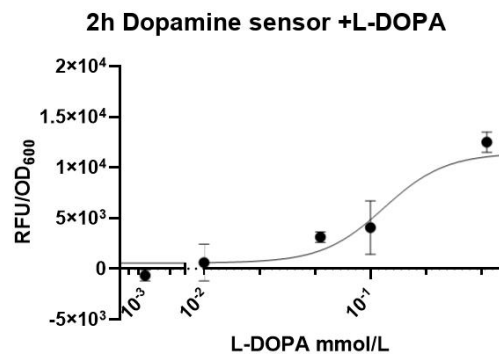


Figure 8. Linear plasmid map from sequencing (correct).



(C) L-DOPA characterization and validation

Conditions: 2 h and 12 h, with and without vitamin C (VC).



(D) Next week's plan

1. Change variables for the L-DOPA sensor and conduct characterization and validation.

Week 14 (8.21-8.28)

Module 1:

(A) Verification of Expression Efficiency

- ## 1. Inoculation

Original strains : BW-pYB1a-eGFP、BW-pYB1a-progo-nov1f、BW-pYB1a-nov1f、
BW-pYB1a-progo、*BW-pYB1a-vdh*、BW-pYB1a-vdh-nox

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-eGFP、BW-pYB1a-progo-nov1f、BW-pYB1a-nov1f、BW-pYB1a-progo

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

Using 4-PG to catalyze the induced enzymes for Isoeugenol production and Isoeugenol to catalyze the induced enzymes for vanillin production, vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

Collect 6 OD units of bacterial cells into a centrifuge tube,

centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Tris-Hcl (PH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

Incubate in a shaker at 30°C for 1h ,3h, 6h, 9h ,12h hours.

4.Prepare mobile phase, standards and filter samples

Preparation of the mobile phase.

The mobile phase was prepared: Methanol: 0.1% acetic acid (ice) = 70:30.Methanol: 0.1% acetic acid (ice) = 55:45.Methanol: 0.1% acetic acid (ice) = 65:35

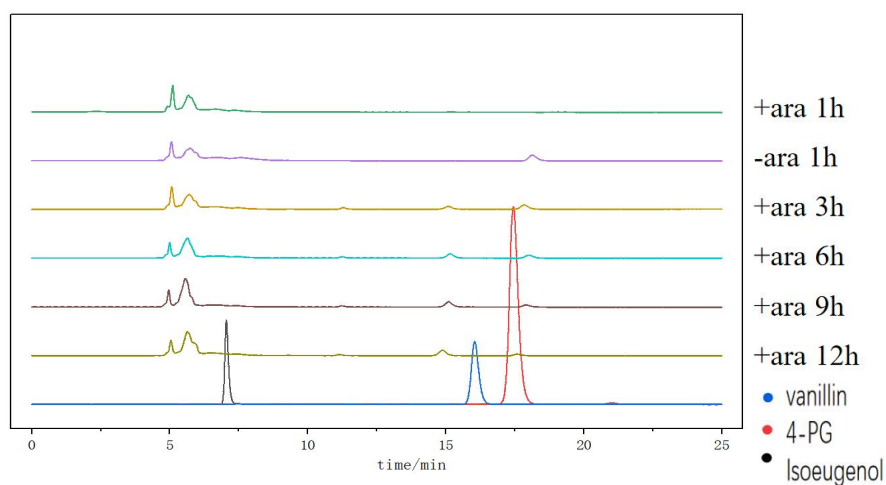
Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22μm membranes for HPLC.

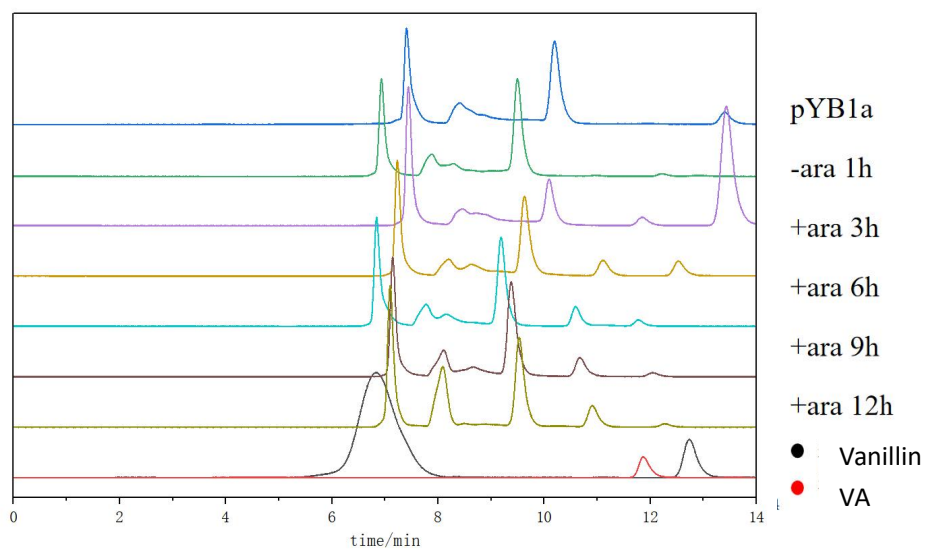
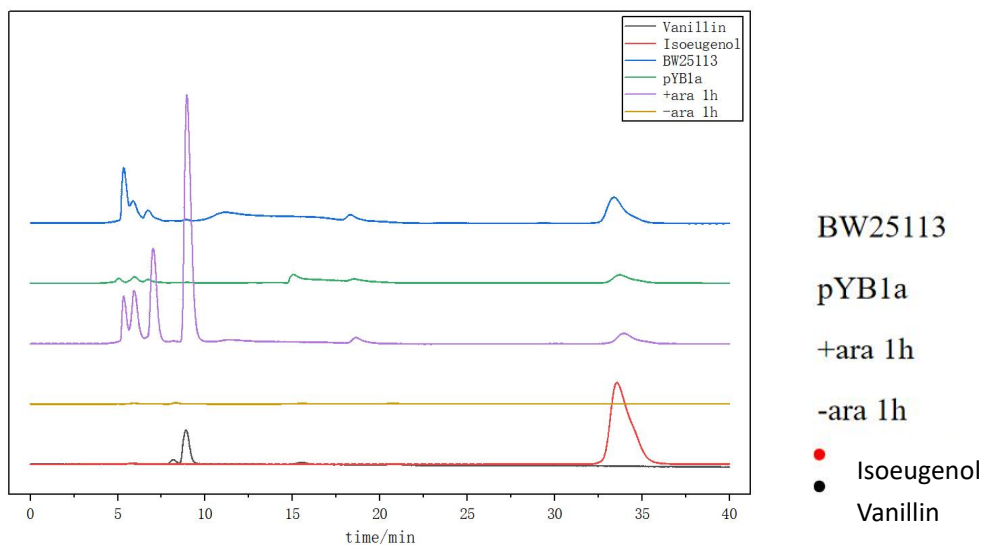
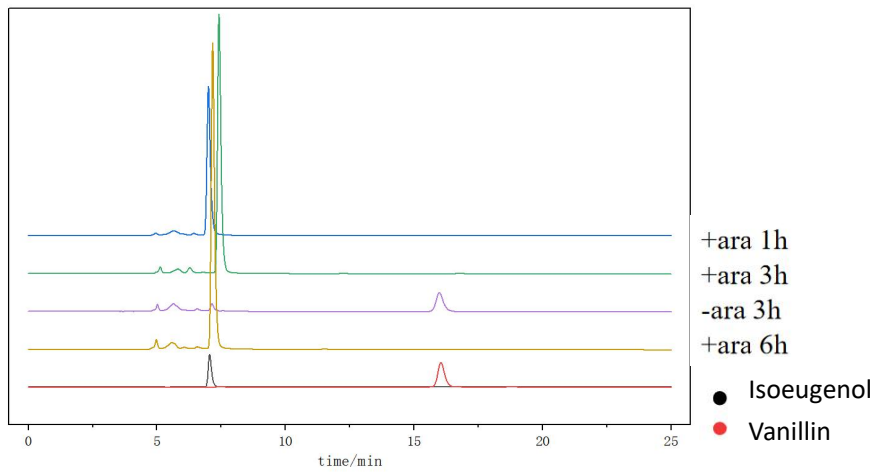
5.HPLC run (sulfuric acid method / phosphoric acid method)

The samples were detected by HPLC using the sulfuric acid method (mobile phase: 5mM H₂SO₄) and the phosphoric acid method. Column: XP-C18; flow rate: 0.4 mL/min、 0.5 mL/min; injection volume: 10μL.

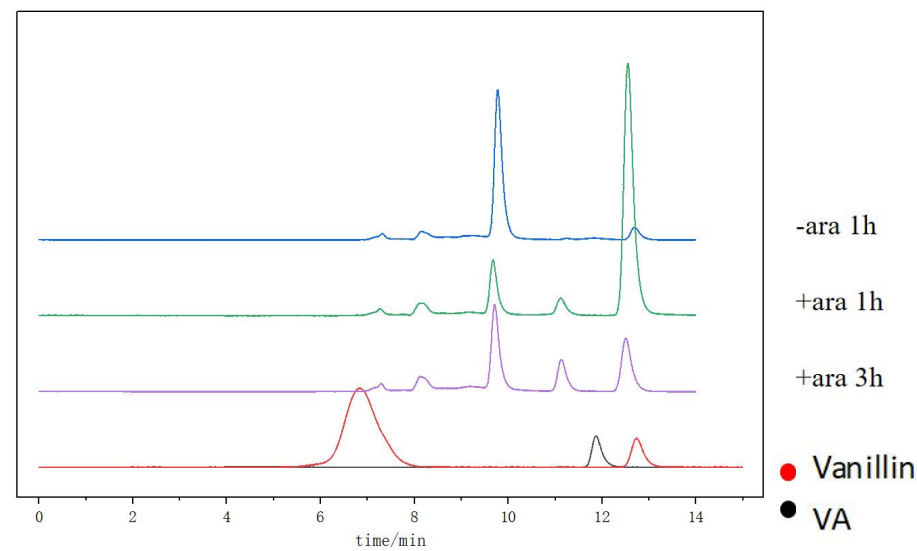
Detection of Isoeugenol



Detection of vanillin



The area of peak 1h is 1236, the area of peak 3h is 2211, the area of peak 6h is 2405, the area of peak 9h is 2659, and the area of peak 12h is 2928. The conversion rate is: 6.24%, the conversion rate of 3h is 11.7%, the conversion rate of 6h is 12.8%, the conversion rate of 9h is 14.2%, and the conversion rate of 12h is 15.7%.



(B)Construct pRB1k-*vdh-nox*

1.PCR+ cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and enzymatic digestion of vector pRB1k

Subsequently, the relevant DNA fragments were assembled using Enzymatic cleavage ligation, followed by transformation into competent cells. Upon completion of the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed.

PCR system (50μL)	
2×Mix	25ul
<i>vdh-nox</i> -F	2ul
<i>vdh-nox</i> -R	2ul
Template	20ng
DDW	To 50ul

PCR system (50μL)	
2×Mix	25ul
<i>KanR-F</i>	2ul
<i>KanR-R</i>	2ul
Template	20ng
DDW	To 50ul

PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	2min45s
72°C	5min
25°C	∞

} × 30

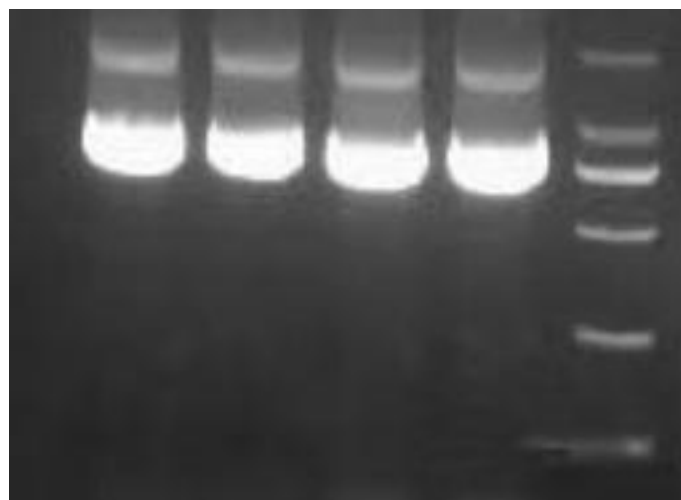
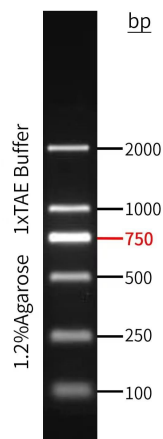
PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	30s
72°C	5min
25°C	∞

} × 30

2.Connection

necting the two fragments using the Enzymatic cleavage ligation,

DL2000 DNA Marker



3.Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



The same replication starting point may result in the loss of fragments.

Module 2:

(A)High-throughput screening of mutant library

1.Screening experiment for the mutant library of BWΔ*codA*-gRB2k-*pobR*^{WT}-P_{pobA}*2-*mCherry-codA*-Cmr/pYB1a-*odm*

VA concentration: 2.5 mM. The 10 mM stock solution was added directly and diluted with water to a final concentration of 2.5 mM.(2) The volume of bacterial suspension was set at 50 μL (1% v/v).(3) Chloramphenicol concentrations: 20 mg/L and 30 mg/L, with a stock concentration of 30 mg/mL.

System(5mL):

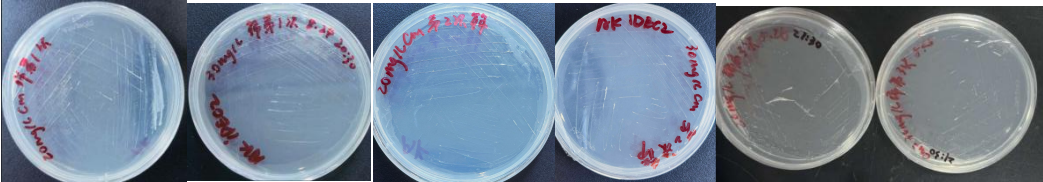
Components	WT(pobR)	mix(AK)	WT(pobR)	mix(AK)
Antibiotics	A5μL、K5μL	A5μL、K5μL	A5μL、K5μL	A5μL、K5μL
5×M9	1mL	1mL	1mL	1mL
20% glycerol	500μL	500μL	500μL	500μL
MgSO ₄	10μL	10μL	10μL	10μL
CaCl ₂	0.5μL	0.5μL	0.5μL	0.5μL
20%ara	50μL	50μL	50μL	50μL
Cm	6.7μL	6.7μL	10μL	10μL
10mM VA	1250μL	1250μL	1250μL	1250μL
DDW	2122.8μL	2122.8μL	2119.5μL	2119.5μL
bacterial suspension	50μL	50μL	50μL	50μL

Change of screening strategy:The mutant library was subcultured when the OD value reached 3–4, and the screening was no longer based on growth comparison with the wild type.

Number of subcultures	OD(20mg/L)	OD(30mg/L)	Growth time
1	5	4.6	42h
2	4.18	3.72	42h
3	0.44	0.23	44h

Problem: The growth time of bacteria was not shortened after subculture.

Streak inoculation for three subculture products: 10 colonies from round 1 and round 2 respectively, 15 colonies from round 3.



1.Induction:

Harvest BW, pYB1a- *eGFP*, pYB1a- *odm* cultured at 37 °C for 12 h. Set non-ara induction control for each group; bring six tubes ZY into clean bench.

Induction system: 50×M 100 μL, 5052 100 μL, MgSO₄ 10 μL, 1000× stock 10 μL, antibiotic 5 μL, bacterial suspension 50 μL, 20% arabinose 50 μL.

Induce at 30 °C with shaking for 16 h.

2. Whole-cell catalysis:

Measure OD₆₀₀ with spectrophotometer, normalize cell density to OD₆₀₀ = 6. Formula: sampling volume (mL) = 6 / measured OD₆₀₀. Aliquot bacteria into 1.5 mL tubes.

Balance tubes, centrifuge 4200 rpm, 10 min; remove all supernatant.

Prepare 200 μ L reaction system: final 2.5 mM vanillic acid in 50 mM Tris- HCl buffer (pH7.5). Add to cell pellet.

Prepare 0 h control for each group, store at -20 °C. Catalysis groups: shake at 30 °C, 200 r/min, dark, 12 h.

3. Sample preparation:

Collect all post-catalysis samples, centrifuge 13000 rpm, 10 min to pellet cells completely.

Carefully aspirate clear supernatant.

Filter supernatant with 0.22 μ m aqueous filter, collect into collection vial.

Load filtrate into new HPLC insert, label, place into reusable HPLC sample vial.

Store all samples at -20 °C prior to HPLC detection.

4. Mobile phase preparation

Prepare mobile phase: 0.1% phosphoric acid aqueous solution : methanol =75:25, mix well.

Prepare 10% methanol and pure methanol, fill into large bottles and label.

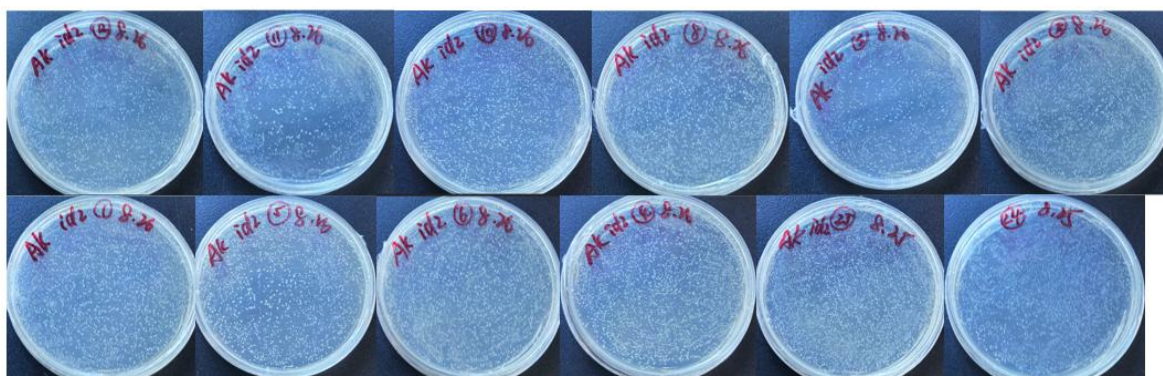
Vacuum-filter aqueous phase into HPLC solvent reservoir to remove fine impurities.

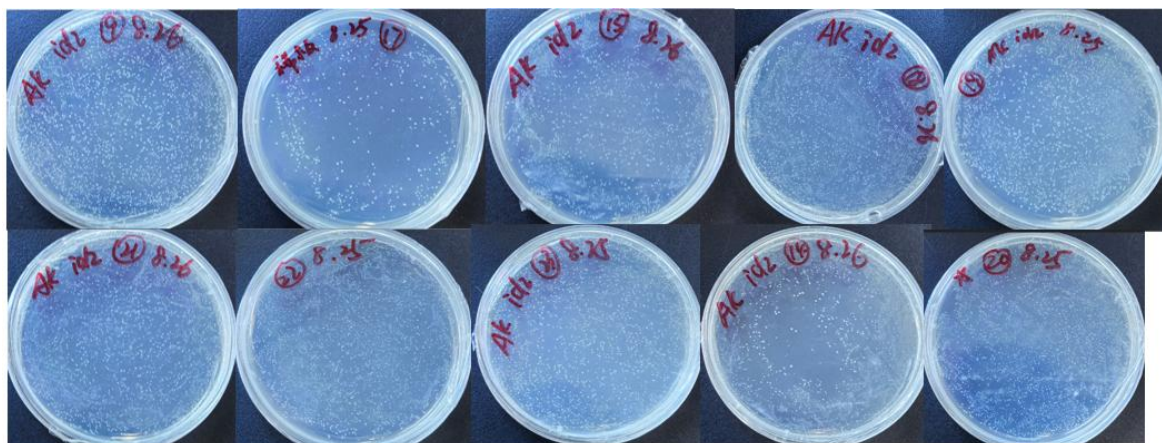
Sonicate for 30 min for degassing to remove dissolved bubbles, avoiding baseline instability and spurious peaks.

(B) Large-scale transformation

1. Transformation

The concentration of mixed plasmids was 20 ng/ μ L. After competent cells were incubated on ice for 5 min, 1 μ L of mixed plasmids was added. The mixture was kept on ice for 30 min, followed by heat shock at 42 °C for 45 s, and then immediately placed on ice for 2–3 min. Subsequently, 600 μ L LB medium was added to the EP tube, and the sample was incubated at 37 °C with shaking for 50 min. The culture was centrifuged at 4200 rpm for 5 min. 100 μ L of the suspension was aspirated, gently mixed, and spread onto agar plates in a clean bench.

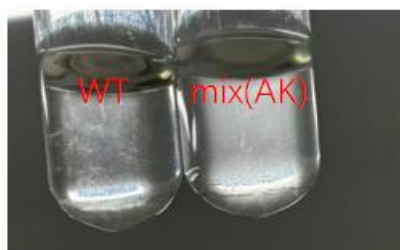




Single concentrated plate: 3000 colonies; library capacity: 66000 clones.

2.Screening assay(8.26)

Results of 22 h screening culture:

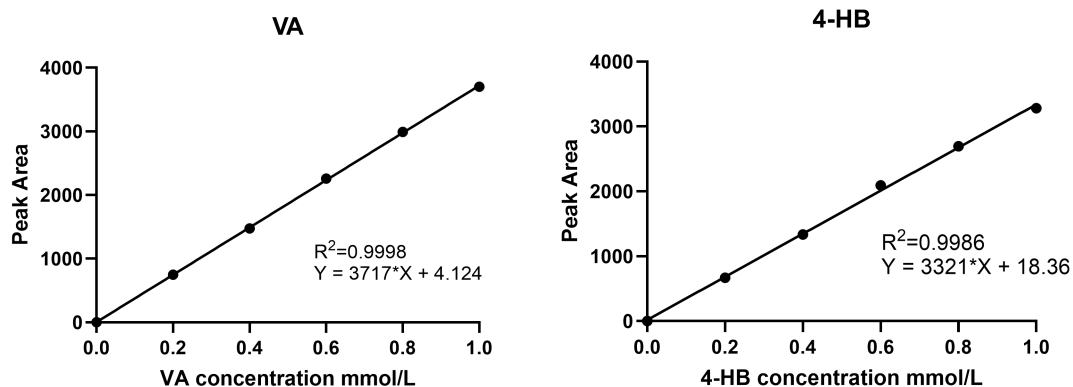


The first round of subculture was completed, and streaking was performed on the products from the first subculture.



(C)HPLC Conditions and Results

1.Standard curve



The standard curve exhibited a good fitting degree. Product concentration of wild-type pYB1a-ODM: 0.003 mmol/L

Conversion rate of wild-type pYB1a-ODM: 0.12%

Module 3:

(A) Pyruvate sensor characterization and validation

1. Loading system

96-well plate, 3 replicate wells per group; upper half 1% inoculum, lower half 1 permille inoculum.

Group	1x M9 medium (uL)	Additional reagent (uL)	Bacterial suspension (uL)
1 BW	198 1x M9	-	2
2 Pyruvate sensor	198 1x M9	-	2
3 Sensor + 1 g/L sodium pyruvate	178 1x M9	1 g/L stock 20	2
4 Sensor + 2 g/L sodium pyruvate	158 1x M9	1 g/L stock 40	2
5 Sensor + 3 g/L sodium pyruvate	138 1x M9	1 g/L stock 60	2
6 Full pathway induced + VA (AS)	196 1x M9 + VA (AS)	Arabinose 2	2
7 Full pathway not induced + VA (AS)	198 1x M9 + VA (AS)	-	2
8 Full pathway induced	198 1x M9 (AS)	Arabinose 2	2
9 Full pathway not induced	198 1x M9 (AS)	-	2
10 eGFP induced	196 1x M9 (A)	Arabinose 2	2
11 eGFP induced + VA	196 1x M9 + VA (A)	Arabinose 2	2

2. Results and analysis

Two rounds of growth curves. Except for the fourth figure, the sensor itself barely grew in the other groups; groups with exogenous sodium pyruvate showed growth, confirming the sensor's effectiveness.

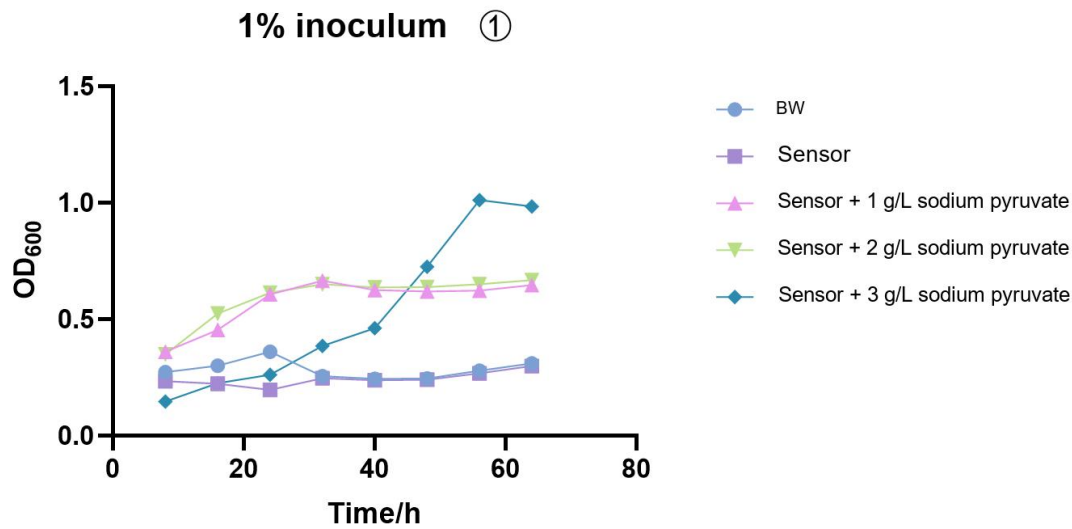


Figure 11. Growth curve at 1% inoculum (round 1).

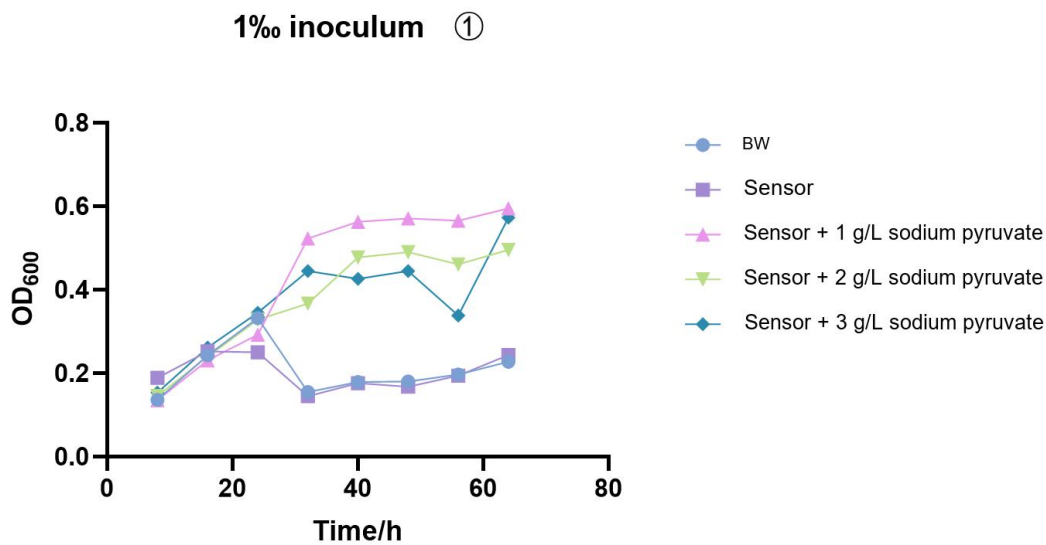


Figure 12. Growth curve at 1‰ inoculum (round 1).

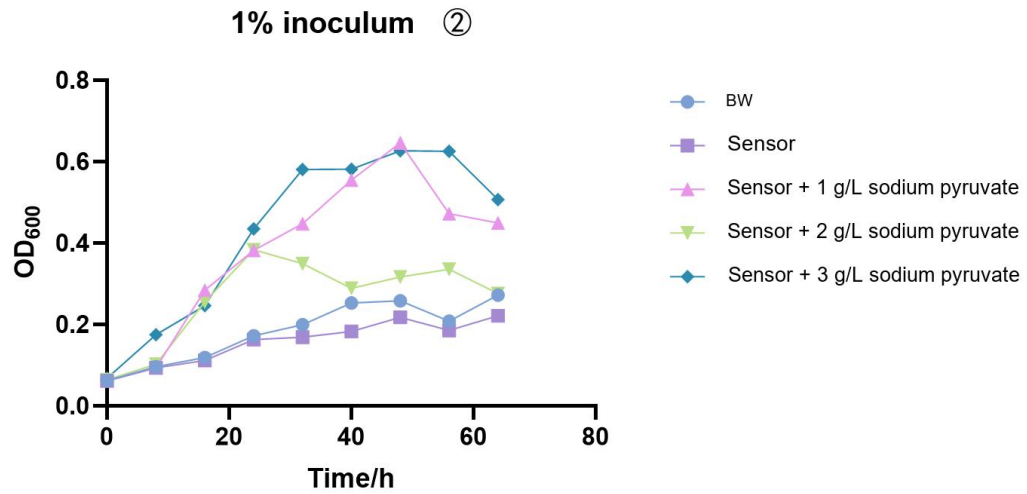


Figure 13. Growth curve at 1% inoculum (round 2).

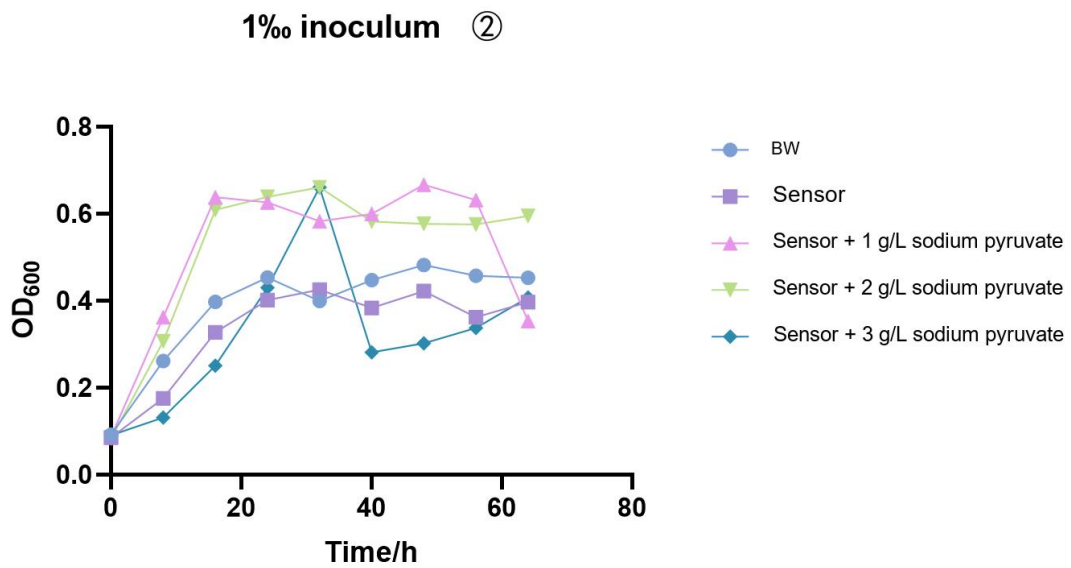


Figure 14. Growth curve at 1 permille inoculum (round 2).

(B) Full pathway verification using the pyruvate sensor

1. Results

Two rounds. First round: all groups failed to grow; figure 3 best matched expectations. Second round: full pathway + VA showed the best growth.

1‰ inoculum ②

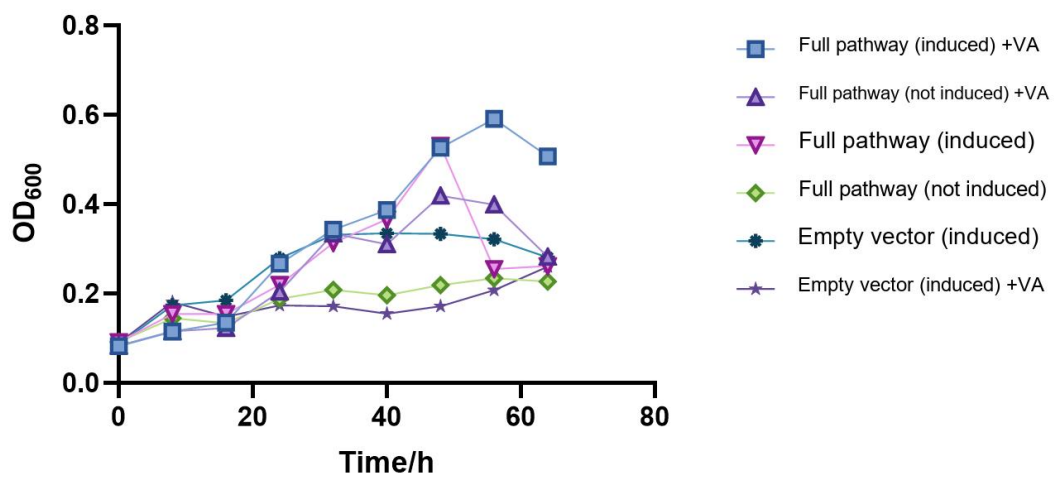


Figure 15. Full-pathway verification at 1 permille inoculum (round 2).

1% inoculum ②

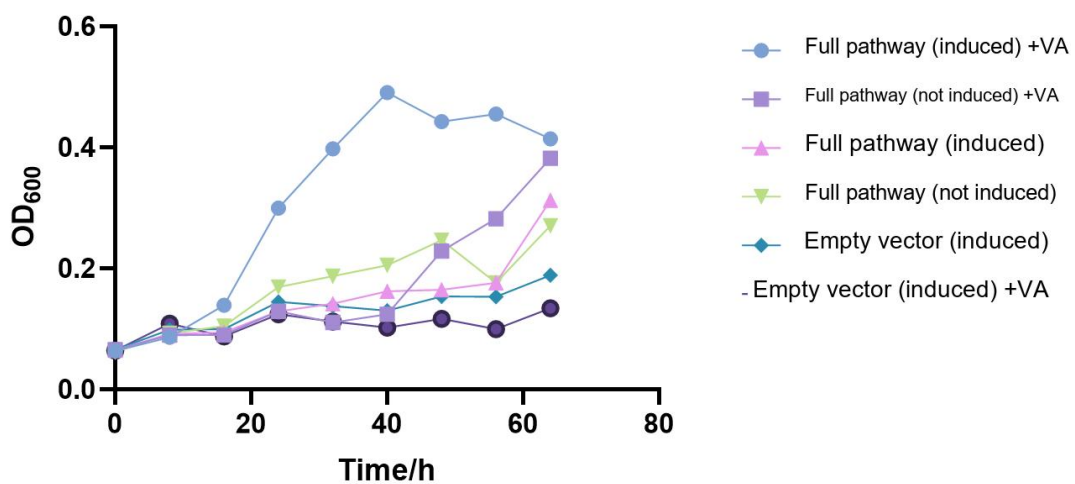


Figure 16. Full-pathway verification at 1% inoculum (round 2).

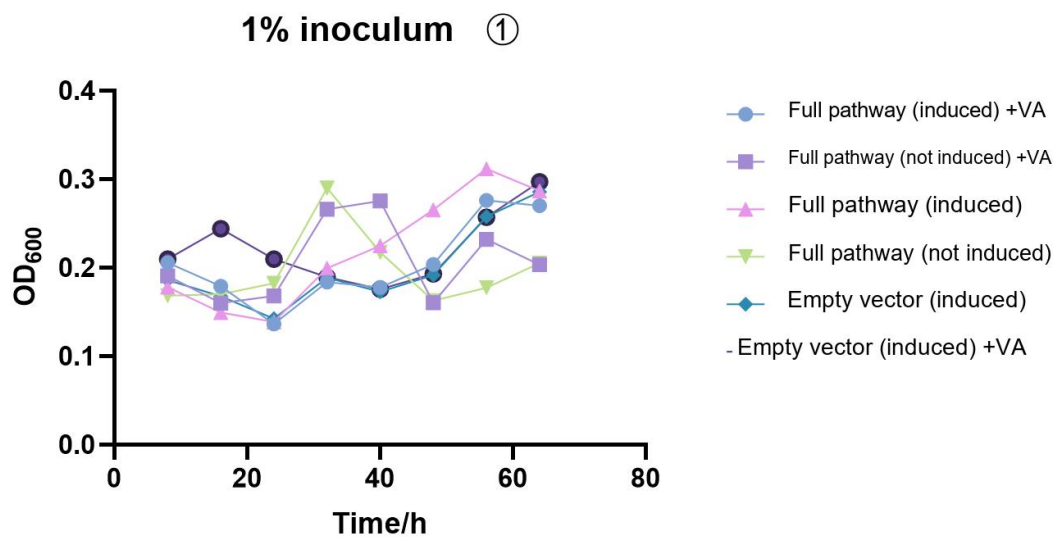


Figure 17. Full-pathway verification at 1% inoculum (round 1).

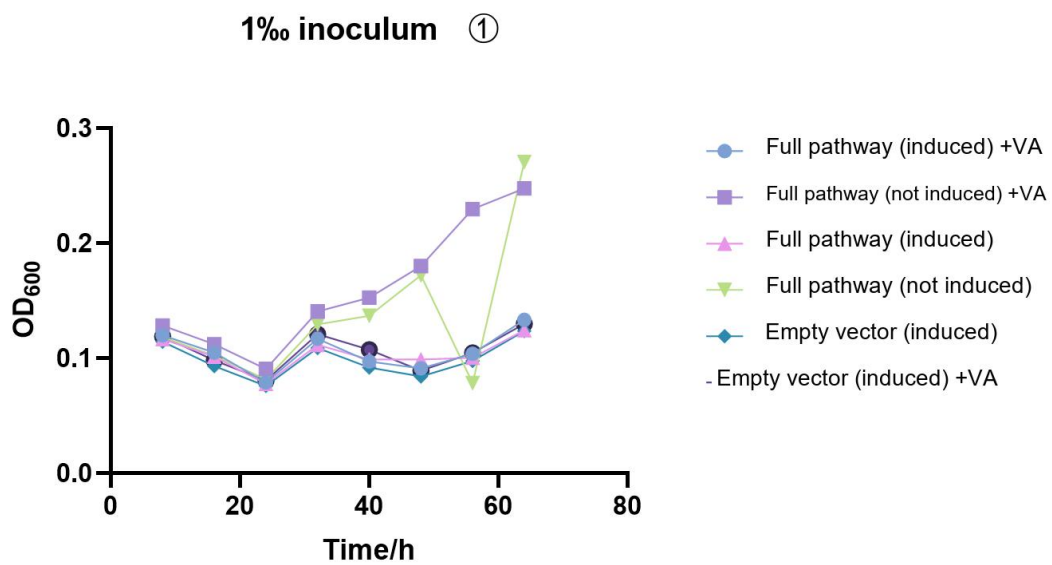


Figure 18. Full-pathway verification at 1 per mille inoculum (round 1).